

Материалы аудиовизуальных отображений, порождаемых СМК-НТТС

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Программа «Симулятор многозадачной координации групп роботизированных наземных транспортно - технологических средств» (СМК-НТТС) генерирует результаты симуляций в форматах *.png, *.csv и *.json для их последующего анализа.

Визуальные материалы типа:

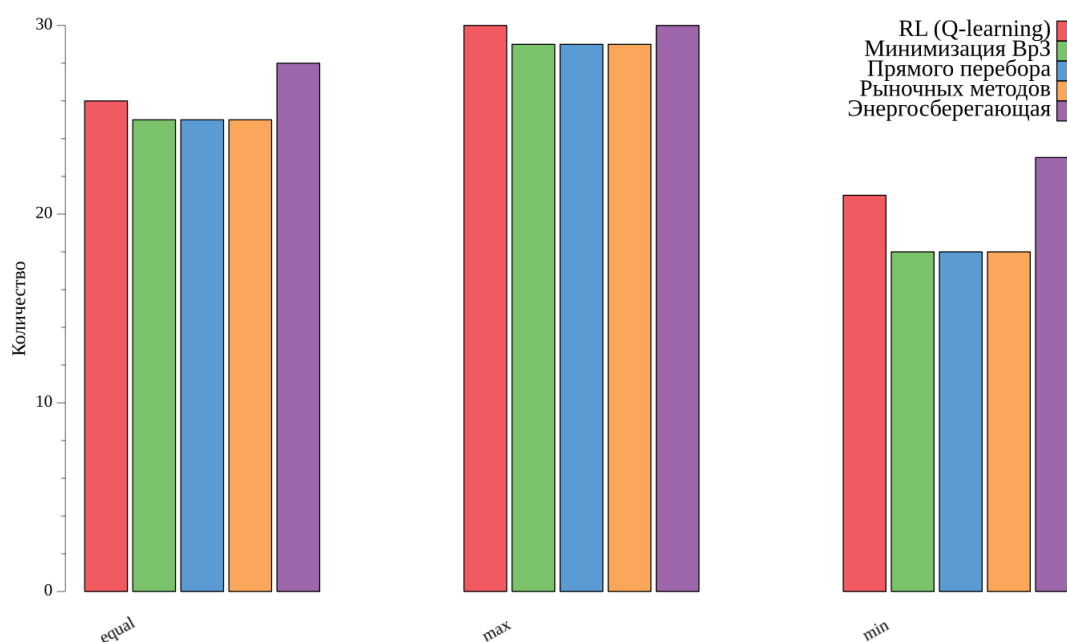


Рис. 1: Выполненные задачи

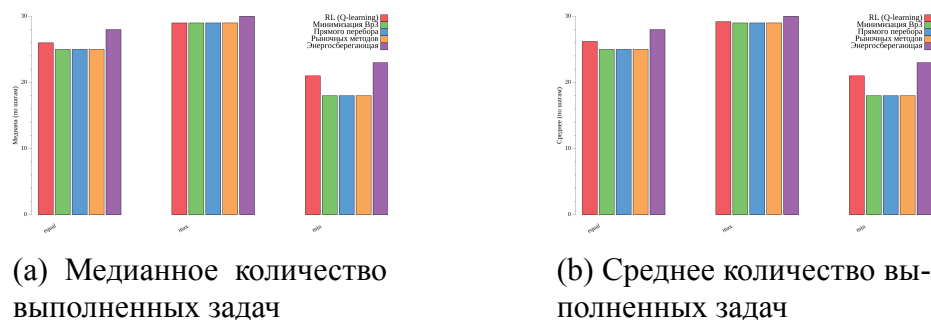


Рис. 2: Сравнительный анализ эффективности стратегий по количеству выполненных задач

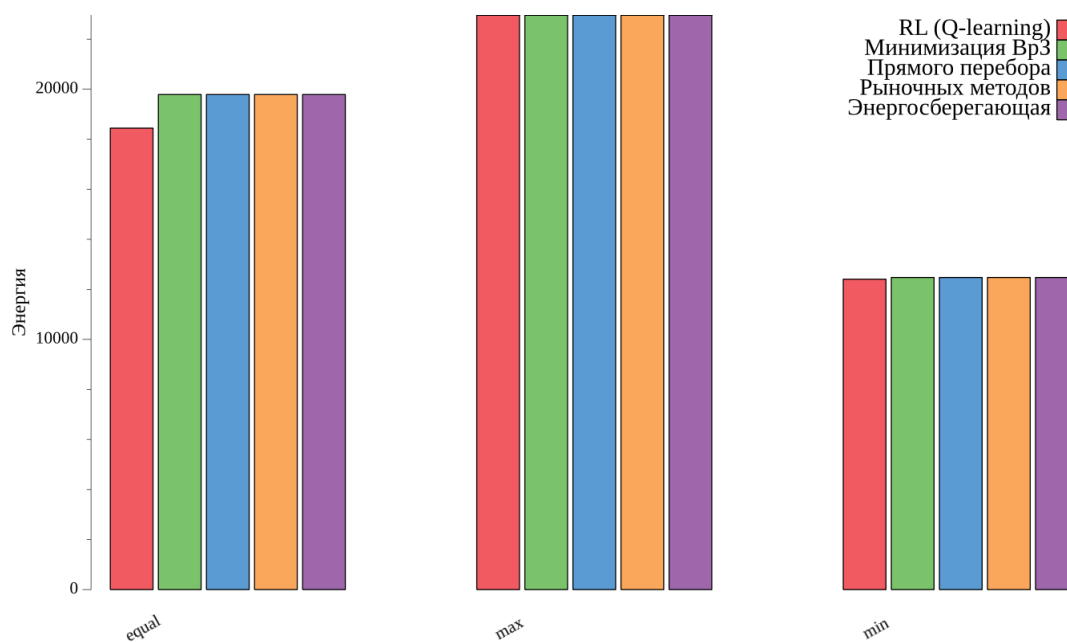


Рис. 3: Потраченная энергия

Печатные материалы типа . A.1 A.2.

А Печатные материалы

А.1 Вывод json

Листинг 1: файл report.json

```
{
  "equal": {
    "RL (Q-learning)": {
      "totalTasks": 32,
      "totalMachine": 8,
      "totalEnergyMachines": 25128,
      "allMachines": [
        {
          "id": "M1",
```

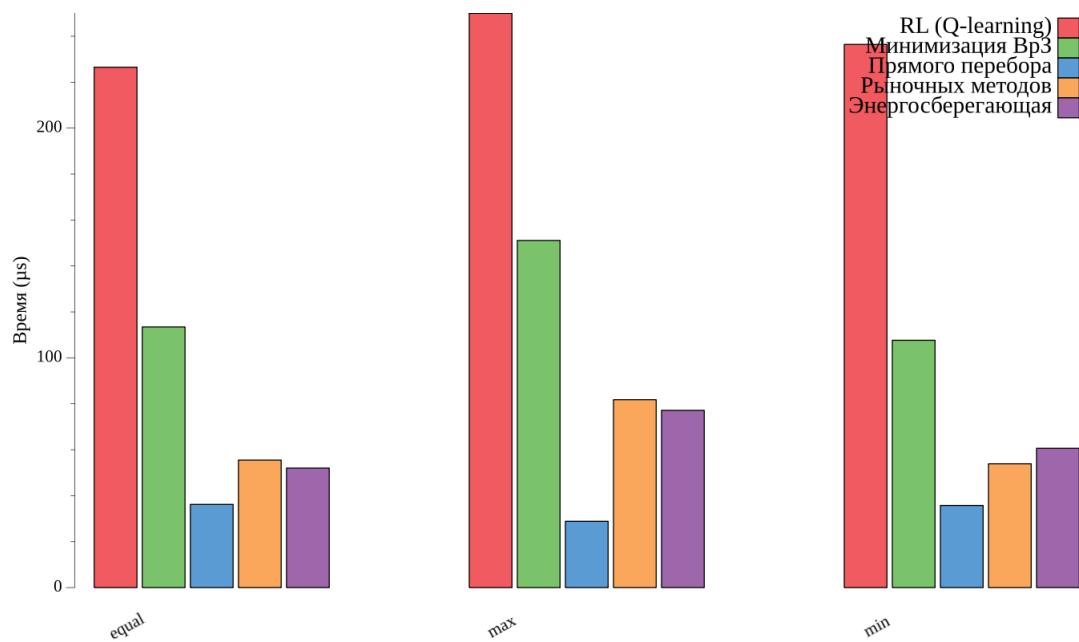


Рис. 4: Общее время выполнения

```

10      "equipment": [
11          1,
12          3,
13          9
14      ],
15      "energy": 3141.75,
16      "AssignedTask": null
17  },
18  {
19      "id": "M2",
20      "equipment": [
21          2,
22          9
23      ],
24      "energy": 3141.75,
25      "AssignedTask": null
26  },
27  {
28      "id": "M3",
29      "equipment": [
30          1,
31          7,
32          14
33      ],
34      "energy": 3141.75,
35      "AssignedTask": null
36  },
37  {
38      "id": "M4",
39      "equipment": [
40          4,

```

```

        5,
        6,
        15
    ],
    "energy": 3141.75,
    "AssignedTask": null
},
{
    "id": "M5",
    "equipment": [
        11
    ],
    "energy": 3141.75,
    "AssignedTask": null
},
{
    "id": "M6",
    "equipment": [
        8,
        13
    ],
    "energy": 3141.75,
    "AssignedTask": null
},
{
    "id": "M7",
    "equipment": [
        10
    ],
    "energy": 3141.75,
    "AssignedTask": null
},
{
    "id": "M8",
    "equipment": [
        12
    ],
    "energy": 3141.75,
    "AssignedTask": null
}
],
"totalEnergyTasks": 25134,
"allTasks": [
    {
        "id": "T23",
        "required_equipment": 8,
        "energy_cost": 1500,
        "CreatedAt": 1759044740
    },
    {
        "id": "T12",
        "required_equipment": 12,

```

```

    "energy_cost": 10,
    "CreatedAt": 1759044729
  },
  {
    "id": "T30",
    "required_equipment": 15,
    "energy_cost": 10,
    "CreatedAt": 1759044747
  },
  {
    "id": "T32",
    "required_equipment": 12,
    "energy_cost": 994,
    "CreatedAt": 1759044719
  },
  {
    "id": "T20",
    "required_equipment": 5,
    "energy_cost": 10,
    "CreatedAt": 1759044737
  },
  {
    "id": "T29",
    "required_equipment": 14,
    "energy_cost": 1500,
    "CreatedAt": 1759044746
  },
  {
    "id": "T25",
    "required_equipment": 10,
    "energy_cost": 1500,
    "CreatedAt": 1759044742
  },
  {
    "id": "T9",
    "required_equipment": 9,
    "energy_cost": 1500,
    "CreatedAt": 1759044726
  },
  {
    "id": "T21",
    "required_equipment": 6,
    "energy_cost": 1500,
    "CreatedAt": 1759044738
  },
  {
    "id": "T28",
    "required_equipment": 13,
    "energy_cost": 10,
    "CreatedAt": 1759044745
  },
  {

```

```

145     "id": "T7",
        "required_equipment": 7,
        "energy_cost": 1500,
        "CreatedAt": 1759044724
150     },
        {
        "id": "T1",
        "required_equipment": 1,
        "energy_cost": 1500,
        "CreatedAt": 1759044718
155     },
        {
        "id": "T10",
        "required_equipment": 10,
        "energy_cost": 10,
160     "CreatedAt": 1759044727
        },
        {
        "id": "T8",
        "required_equipment": 8,
165     "energy_cost": 10,
        "CreatedAt": 1759044725
        },
        {
        "id": "T14",
170     "required_equipment": 14,
        "energy_cost": 10,
        "CreatedAt": 1759044731
        },
        {
175     "id": "T18",
        "required_equipment": 3,
        "energy_cost": 10,
        "CreatedAt": 1759044735
        },
180     {
        "id": "T26",
        "required_equipment": 11,
        "energy_cost": 10,
        "CreatedAt": 1759044743
185     },
        {
        "id": "T2",
        "required_equipment": 2,
        "energy_cost": 10,
190     "CreatedAt": 1759044719
        },
        {
        "id": "T4",
195     "required_equipment": 4,
        "energy_cost": 10,
        "CreatedAt": 1759044721

```

```
    },
    {
200       "id": "T3",
       "required_equipment": 3,
       "energy_cost": 1500,
       "CreatedAt": 1759044720
    },
    {
205       "id": "T11",
       "required_equipment": 11,
       "energy_cost": 1500,
       "CreatedAt": 1759044728
    },
210     {
       "id": "T16",
       "required_equipment": 1,
       "energy_cost": 10,
       "CreatedAt": 1759044733
215     },
    {
       "id": "T22",
       "required_equipment": 7,
       "energy_cost": 10,
220       "CreatedAt": 1759044739
    },
    {
       "id": "T17",
       "required_equipment": 2,
225       "energy_cost": 1500,
       "CreatedAt": 1759044734
    },
    {
       "id": "T24",
230       "required_equipment": 9,
       "energy_cost": 10,
       "CreatedAt": 1759044741
    },
    {
235       "id": "T19",
       "required_equipment": 4,
       "energy_cost": 1500,
       "CreatedAt": 1759044736
    },
240     {
       "id": "T27",
       "required_equipment": 12,
       "energy_cost": 1500,
       "CreatedAt": 1759044744
245     },
    {
       "id": "T13",
       "required_equipment": 13,
```

```

250         "energy_cost": 1500,
        "CreatedAt": 1759044730
    },
    {
        "id": "T31",
        "required_equipment": 7,
255        "energy_cost": 1490,
        "CreatedAt": 1759044718
    },
    {
        "id": "T15",
260        "required_equipment": 15,
        "energy_cost": 1500,
        "CreatedAt": 1759044732
    },
    {
265        "id": "T5",
        "required_equipment": 5,
        "energy_cost": 1500,
        "CreatedAt": 1759044722
    },
270    {
        "id": "T6",
        "required_equipment": 6,
        "energy_cost": 10,
        "CreatedAt": 1759044723
275    }
],
"totalTime": "206.021µs",
"stepMetrics": [
    {
280        "step": 1,
        "lenTasksDone": 28,
        "tasksDone": [
            "T16",
            "T24",
285            "T22",
            "T30",
            "T26",
            "T28",
            "T10",
290            "T12",
            "T1",
            "T9",
            "T14",
            "T15",
            "T11",
295            "T8",
            "T25",
            "T32",
            "T18",
            "T2",
300

```



```

305         "T29",
            "T20",
            "T23",
            "T27",
            "T3",
            "T17",
            "T31",
            "T5",
            "T13",
310         "T6"
        ],
        "energyUsed": 20867.5,
        "lenNotExecTasks": 4,
        "realTime": "27.032µs"
315    },
    {
        "step": 2,
        "lenTasksDone": 27,
        "tasksDone": [
320         "T16",
            "T2",
            "T1",
            "T20",
            "T26",
325         "T8",
            "T10",
            "T12",
            "T18",
            "T17",
330         "T14",
            "T6",
            "T11",
            "T23",
            "T25",
335         "T32",
            "T3",
            "T24",
            "T29",
            "T5",
340         "T28",
            "T27",
            "T9",
            "T22",
            "T21",
345         "T13",
            "T30"
        ],
        "energyUsed": 19367.5,
        "lenNotExecTasks": 5,
        "realTime": "29.611µs"
350    },
    {

```

```

355     "step": 3,
        "lenTasksDone": 28,
        "tasksDone": [
            "T18",
            "T24",
            "T14",
            "T20",
360     "T26",
            "T8",
            "T10",
            "T12",
            "T16",
365     "T9",
            "T29",
            "T5",
            "T11",
            "T23",
370     "T25",
            "T32",
            "T1",
            "T2",
            "T22",
375     "T30",
            "T28",
            "T27",
            "T3",
            "T17",
380     "T31",
            "T15",
            "T13",
            "T6"
        ],
385     "energyUsed": 20867.5,
        "lenNotExecTasks": 4,
        "realTime": "80.557μs"
    },
    {
390     "step": 4,
        "lenTasksDone": 28,
        "tasksDone": [
            "T16",
            "T24",
395     "T14",
            "T20",
            "T26",
            "T8",
            "T10",
            "T12",
400     "T1",
            "T9",
            "T29",
            "T5",

```

```

405         "T11",
            "T23",
            "T25",
            "T32",
            "T18",
410         "T2",
            "T22",
            "T30",
            "T28",
            "T27",
415         "T3",
            "T17",
            "T31",
            "T15",
            "T13",
420         "T6"
    ],
    "energyUsed": 20867.5,
    "lenNotExecTasks": 4,
    "realTime": "32.032μs"
425 },
    {
        "step": 5,
        "lenTasksDone": 27,
        "tasksDone": [
430         "T16",
            "T24",
            "T14",
            "T20",
            "T26",
435         "T8",
            "T10",
            "T12",
            "T1",
            "T2",
440         "T29",
            "T30",
            "T11",
            "T23",
            "T25",
445         "T32",
            "T9",
            "T17",
            "T22",
            "T5",
450         "T28",
            "T27",
            "T18",
            "T31",
            "T15",
455         "T13",
            "T6"
        ]
    }

```

```

    ],
    "energyUsed": 19489.25,
    "lenNotExecTasks": 5,
    "realTime": "29.934µs"
  }
]
},
Минимизация" Вр3": {
  "totalTasks": 32,
  "totalMachine": 8,
  "totalEnergyMachines": 25128,
  "allMachines": [
    {
      "id": "M1",
      "equipment": [
        1,
        3,
        9
      ],
      "energy": 3141.75,
      "AssignedTask": null
    },
    {
      "id": "M2",
      "equipment": [
        2,
        9
      ],
      "energy": 3141.75,
      "AssignedTask": null
    },
    {
      "id": "M3",
      "equipment": [
        1,
        7,
        14
      ],
      "energy": 3141.75,
      "AssignedTask": null
    },
    {
      "id": "M4",
      "equipment": [
        4,
        5,
        6,
        15
      ],
      "energy": 3141.75,
      "AssignedTask": null
    },
  ],

```

```

510     {
        "id": "M5",
        "equipment": [
            11
        ],
        "energy": 3141.75,
515     "AssignedTask": null
    },
    {
        "id": "M6",
        "equipment": [
520         8,
            13
        ],
        "energy": 3141.75,
        "AssignedTask": null
525     },
    {
        "id": "M7",
        "equipment": [
            10
530         ],
        "energy": 3141.75,
        "AssignedTask": null
    },
    {
535     "id": "M8",
        "equipment": [
            12
        ],
        "energy": 3141.75,
540     "AssignedTask": null
    }
],
"totalEnergyTasks": 25134,
"allTasks": [
545     {
        "id": "T23",
        "required_equipment": 8,
        "energy_cost": 1500,
        "CreatedAt": 1759044740
550     },
    {
        "id": "T12",
        "required_equipment": 12,
        "energy_cost": 10,
555     "CreatedAt": 1759044729
    },
    {
        "id": "T30",
        "required_equipment": 15,
560     "energy_cost": 10,

```

```

    "CreatedAt": 1759044747
  },
  {
    "id": "T32",
    "required_equipment": 12,
    "energy_cost": 994,
    "CreatedAt": 1759044719
  },
  {
    "id": "T20",
    "required_equipment": 5,
    "energy_cost": 10,
    "CreatedAt": 1759044737
  },
  {
    "id": "T29",
    "required_equipment": 14,
    "energy_cost": 1500,
    "CreatedAt": 1759044746
  },
  {
    "id": "T25",
    "required_equipment": 10,
    "energy_cost": 1500,
    "CreatedAt": 1759044742
  },
  {
    "id": "T9",
    "required_equipment": 9,
    "energy_cost": 1500,
    "CreatedAt": 1759044726
  },
  {
    "id": "T21",
    "required_equipment": 6,
    "energy_cost": 1500,
    "CreatedAt": 1759044738
  },
  {
    "id": "T28",
    "required_equipment": 13,
    "energy_cost": 10,
    "CreatedAt": 1759044745
  },
  {
    "id": "T7",
    "required_equipment": 7,
    "energy_cost": 1500,
    "CreatedAt": 1759044724
  },
  {
    "id": "T1",

```

```

        "required_equipment": 1,
        "energy_cost": 1500,
615      "CreatedAt": 1759044718
    },
    {
        "id": "T10",
        "required_equipment": 10,
620      "energy_cost": 10,
        "CreatedAt": 1759044727
    },
    {
        "id": "T8",
625      "required_equipment": 8,
        "energy_cost": 10,
        "CreatedAt": 1759044725
    },
    {
630      "id": "T14",
        "required_equipment": 14,
        "energy_cost": 10,
        "CreatedAt": 1759044731
    },
635    {
        "id": "T18",
        "required_equipment": 3,
        "energy_cost": 10,
        "CreatedAt": 1759044735
640    },
    {
        "id": "T26",
        "required_equipment": 11,
645      "energy_cost": 10,
        "CreatedAt": 1759044743
    },
    {
        "id": "T2",
650      "required_equipment": 2,
        "energy_cost": 10,
        "CreatedAt": 1759044719
    },
    {
655      "id": "T4",
        "required_equipment": 4,
        "energy_cost": 10,
        "CreatedAt": 1759044721
    },
660    {
        "id": "T3",
        "required_equipment": 3,
        "energy_cost": 1500,
        "CreatedAt": 1759044720
    },
    },

```

```

665     {
        "id": "T11",
        "required_equipment": 11,
        "energy_cost": 1500,
        "CreatedAt": 1759044728
670     },
        {
        "id": "T16",
        "required_equipment": 1,
        "energy_cost": 10,
675     "CreatedAt": 1759044733
        },
        {
        "id": "T22",
        "required_equipment": 7,
680     "energy_cost": 10,
        "CreatedAt": 1759044739
        },
        {
        "id": "T17",
685     "required_equipment": 2,
        "energy_cost": 1500,
        "CreatedAt": 1759044734
        },
        {
690     "id": "T24",
        "required_equipment": 9,
        "energy_cost": 10,
        "CreatedAt": 1759044741
        },
695     {
        "id": "T19",
        "required_equipment": 4,
        "energy_cost": 1500,
        "CreatedAt": 1759044736
700     },
        {
        "id": "T27",
        "required_equipment": 12,
        "energy_cost": 1500,
705     "CreatedAt": 1759044744
        },
        {
        "id": "T13",
        "required_equipment": 13,
710     "energy_cost": 1500,
        "CreatedAt": 1759044730
        },
        {
715     "id": "T31",
        "required_equipment": 7,
        "energy_cost": 1490,

```



```

    "CreatedAt": 1759044718
  },
  {
    "id": "T15",
    "required_equipment": 15,
    "energy_cost": 1500,
    "CreatedAt": 1759044732
  },
  {
    "id": "T5",
    "required_equipment": 5,
    "energy_cost": 1500,
    "CreatedAt": 1759044722
  },
  {
    "id": "T6",
    "required_equipment": 6,
    "energy_cost": 10,
    "CreatedAt": 1759044723
  }
],
"totalTime": "85.674µs",
"stepMetrics": [
  {
    "step": 1,
    "lenTasksDone": 27,
    "tasksDone": [
      "T23",
      "T12",
      "T30",
      "T29",
      "T25",
      "T9",
      "T26",
      "T32",
      "T20",
      "T28",
      "T7",
      "T1",
      "T10",
      "T2",
      "T11",
      "T21",
      "T8",
      "T14",
      "T18",
      "T17",
      "T27",
      "T4",
      "T16",
      "T24",
      "T13",

```

```

770         "T22",
           "T19"
       ],
       "energyUsed": 19489.25,
       "lenNotExecTasks": 5,
       "realTime": "29.62µs"
775   },
   {
       "step": 2,
       "lenTasksDone": 27,
       "tasksDone": [
780         "T23",
           "T12",
           "T30",
           "T29",
           "T25",
785         "T9",
           "T26",
           "T32",
           "T20",
           "T28",
790         "T7",
           "T1",
           "T10",
           "T2",
           "T11",
795         "T21",
           "T8",
           "T14",
           "T18",
           "T17",
800         "T27",
           "T4",
           "T16",
           "T24",
           "T13",
805         "T22",
           "T19"
       ],
       "energyUsed": 19489.25,
       "lenNotExecTasks": 5,
       "realTime": "16.461µs"
810   },
   {
       "step": 3,
       "lenTasksDone": 27,
       "tasksDone": [
815         "T23",
           "T12",
           "T30",
           "T29",
           "T25",
820         "T23",
           "T12",
           "T30",
           "T29",
           "T25",

```

```

      "T9",
      "T26",
      "T32",
      "T20",
825      "T28",
      "T7",
      "T1",
      "T10",
      "T2",
830      "T11",
      "T21",
      "T8",
      "T14",
      "T18",
835      "T17",
      "T27",
      "T4",
      "T16",
      "T24",
840      "T13",
      "T22",
      "T19"
    ],
    "energyUsed": 19489.25,
845    "lenNotExecTasks": 5,
    "realTime": "12.494µs"
  },
  {
    "step": 4,
850    "lenTasksDone": 27,
    "tasksDone": [
      "T23",
      "T12",
      "T30",
855      "T29",
      "T25",
      "T9",
      "T26",
      "T32",
860      "T20",
      "T28",
      "T7",
      "T1",
      "T10",
865      "T2",
      "T11",
      "T21",
      "T8",
      "T14",
870      "T18",
      "T17",
      "T27",

```

```

875         "T4",
            "T16",
            "T24",
            "T13",
            "T22",
            "T19"
880     ],
        "energyUsed": 19489.25,
        "lenNotExecTasks": 5,
        "realTime": "12.675µs"
    },
    {
885        "step": 5,
        "lenTasksDone": 27,
        "tasksDone": [
            "T23",
            "T12",
890            "T30",
            "T29",
            "T25",
            "T9",
            "T26",
            "T32",
895            "T20",
            "T28",
            "T7",
            "T1",
            "T10",
            "T2",
            "T11",
            "T21",
            "T8",
            "T14",
900            "T18",
            "T17",
            "T27",
            "T4",
            "T16",
            "T24",
            "T13",
            "T22",
            "T19"
905        ],
        "energyUsed": 19489.25,
        "lenNotExecTasks": 5,
        "realTime": "12.018µs"
910    }
]
915 },
Прямого" перебора": {
    "totalTasks": 32,
    "totalMachine": 8,
920

```

```

925 "totalEnergyMachines": 25128,
    "allMachines": [
      {
        "id": "M1",
        "equipment": [
930     1,
        3,
        9
        ],
        "energy": 3141.75,
935     "AssignedTask": null
      },
      {
        "id": "M2",
        "equipment": [
940     2,
        9
        ],
        "energy": 3141.75,
        "AssignedTask": null
945     },
      {
        "id": "M3",
        "equipment": [
950     1,
        7,
        14
        ],
        "energy": 3141.75,
        "AssignedTask": null
955     },
      {
        "id": "M4",
        "equipment": [
960     4,
        5,
        6,
        15
        ],
        "energy": 3141.75,
965     "AssignedTask": null
      },
      {
        "id": "M5",
        "equipment": [
970     11
        ],
        "energy": 3141.75,
        "AssignedTask": null
975     },
      {
        "id": "M6",

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    "equipment": [
      8,
      13
    ],
    "energy": 3141.75,
    "AssignedTask": null
  },
  {
    "id": "M7",
    "equipment": [
      10
    ],
    "energy": 3141.75,
    "AssignedTask": null
  },
  {
    "id": "M8",
    "equipment": [
      12
    ],
    "energy": 3141.75,
    "AssignedTask": null
  }
],
"totalEnergyTasks": 25134,
"allTasks": [
  {
    "id": "T23",
    "required_equipment": 8,
    "energy_cost": 1500,
    "CreatedAt": 1759044740
  },
  {
    "id": "T12",
    "required_equipment": 12,
    "energy_cost": 10,
    "CreatedAt": 1759044729
  },
  {
    "id": "T30",
    "required_equipment": 15,
    "energy_cost": 10,
    "CreatedAt": 1759044747
  },
  {
    "id": "T32",
    "required_equipment": 12,
    "energy_cost": 994,
    "CreatedAt": 1759044719
  },
  {
    "id": "T20",

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1030         "required_equipment": 5,
           "energy_cost": 10,
           "CreatedAt": 1759044737
1035     },
           {
           "id": "T29",
           "required_equipment": 14,
           "energy_cost": 1500,
           "CreatedAt": 1759044746
1040     },
           {
           "id": "T25",
           "required_equipment": 10,
           "energy_cost": 1500,
           "CreatedAt": 1759044742
1045     },
           {
           "id": "T9",
           "required_equipment": 9,
           "energy_cost": 1500,
           "CreatedAt": 1759044726
1050     },
           {
           "id": "T21",
           "required_equipment": 6,
           "energy_cost": 1500,
           "CreatedAt": 1759044738
1055     },
           {
           "id": "T28",
           "required_equipment": 13,
           "energy_cost": 10,
           "CreatedAt": 1759044745
1060     },
           {
           "id": "T7",
           "required_equipment": 7,
           "energy_cost": 1500,
           "CreatedAt": 1759044724
1065     },
           {
           "id": "T1",
           "required_equipment": 1,
           "energy_cost": 1500,
           "CreatedAt": 1759044718
1070     },
           {
           "id": "T10",
           "required_equipment": 10,
           "energy_cost": 10,
           "CreatedAt": 1759044727
1075     },
           {
1080     },

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1085     {
        "id": "T8",
        "required_equipment": 8,
        "energy_cost": 10,
        "CreatedAt": 1759044725
    },
1090     {
        "id": "T14",
        "required_equipment": 14,
        "energy_cost": 10,
        "CreatedAt": 1759044731
    },
1095     {
        "id": "T18",
        "required_equipment": 3,
        "energy_cost": 10,
        "CreatedAt": 1759044735
    },
1100     {
        "id": "T26",
        "required_equipment": 11,
        "energy_cost": 10,
        "CreatedAt": 1759044743
    },
1105     {
        "id": "T2",
        "required_equipment": 2,
        "energy_cost": 10,
        "CreatedAt": 1759044719
1110     },
    {
        "id": "T4",
        "required_equipment": 4,
        "energy_cost": 10,
1115     "CreatedAt": 1759044721
    },
    {
        "id": "T3",
        "required_equipment": 3,
1120     "energy_cost": 1500,
        "CreatedAt": 1759044720
    },
    {
        "id": "T11",
1125     "required_equipment": 11,
        "energy_cost": 1500,
        "CreatedAt": 1759044728
    },
1130     {
        "id": "T16",
        "required_equipment": 1,
        "energy_cost": 10,

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    "CreatedAt": 1759044733
  },
1135 {
    "id": "T22",
    "required_equipment": 7,
    "energy_cost": 10,
    "CreatedAt": 1759044739
1140 },
  {
    "id": "T17",
    "required_equipment": 2,
    "energy_cost": 1500,
1145 "CreatedAt": 1759044734
  },
  {
    "id": "T24",
    "required_equipment": 9,
1150 "energy_cost": 10,
    "CreatedAt": 1759044741
  },
  {
    "id": "T19",
    "required_equipment": 4,
1155 "energy_cost": 1500,
    "CreatedAt": 1759044736
  },
  {
1160 "id": "T27",
    "required_equipment": 12,
    "energy_cost": 1500,
    "CreatedAt": 1759044744
  },
1165 {
    "id": "T13",
    "required_equipment": 13,
    "energy_cost": 1500,
    "CreatedAt": 1759044730
1170 },
  {
    "id": "T31",
    "required_equipment": 7,
    "energy_cost": 1490,
1175 "CreatedAt": 1759044718
  },
  {
    "id": "T15",
    "required_equipment": 15,
1180 "energy_cost": 1500,
    "CreatedAt": 1759044732
  },
  {
    "id": "T5",

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1185         "required_equipment": 5,
            "energy_cost": 1500,
            "CreatedAt": 1759044722
        },
        {
1190         "id": "T6",
            "required_equipment": 6,
            "energy_cost": 10,
            "CreatedAt": 1759044723
        }
1195     ],
    "totalTime": "17.526µs",
    "stepMetrics": [
        {
1200         "step": 1,
            "lenTasksDone": 27,
            "tasksDone": [
                "T9",
                "T2",
                "T29",
1205         "T30",
                "T26",
                "T23",
                "T25",
                "T12",
1210         "T1",
                "T17",
                "T7",
                "T20",
                "T11",
1215         "T28",
                "T10",
                "T32",
                "T18",
                "T24",
1220         "T14",
                "T21",
                "T8",
                "T27",
                "T16",
1225         "T4",
                "T13",
                "T22",
                "T19"
            ],
1230         "energyUsed": 19489.25,
            "lenNotExecTasks": 5,
            "realTime": "3.171µs"
        },
        {
1235         "step": 2,
            "lenTasksDone": 27,

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1240     "tasksDone": [
        "T9",
        "T2",
        "T29",
        "T30",
        "T26",
        "T23",
        "T25",
1245     "T12",
        "T1",
        "T17",
        "T7",
        "T20",
1250     "T11",
        "T28",
        "T10",
        "T32",
        "T18",
1255     "T24",
        "T14",
        "T21",
        "T8",
        "T27",
1260     "T16",
        "T4",
        "T13",
        "T22",
        "T19"
1265     ],
        "energyUsed": 19489.25,
        "lenNotExecTasks": 5,
        "realTime": "2.035µs"
1270     },
    {
        "step": 3,
        "lenTasksDone": 27,
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            "T2",
            "T29",
            "T30",
            "T26",
            "T23",
            "T25",
1280     "T12",
            "T1",
            "T17",
            "T7",
            "T20",
1285     "T11",
            "T28",
            "T10",

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1290         "T32",
        "T18",
        "T24",
        "T14",
        "T21",
1295         "T8",
        "T27",
        "T16",
        "T4",
        "T13",
        "T22",
1300         "T19"
    ],
    "energyUsed": 19489.25,
    "lenNotExecTasks": 5,
    "realTime": "1.562µs"
1305 },
{
    "step": 4,
    "lenTasksDone": 27,
    "tasksDone": [
1310         "T9",
        "T2",
        "T29",
        "T30",
        "T26",
1315         "T23",
        "T25",
        "T12",
        "T1",
        "T17",
1320         "T7",
        "T20",
        "T11",
        "T28",
        "T10",
1325         "T32",
        "T18",
        "T24",
        "T14",
        "T21",
1330         "T8",
        "T27",
        "T16",
        "T4",
        "T13",
1335         "T22",
        "T19"
    ],
    "energyUsed": 19489.25,
    "lenNotExecTasks": 5,
1340     "realTime": "6.458µs"

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    },
    {
      "step": 5,
      "lenTasksDone": 27,
1345     "tasksDone": [
          "T9",
          "T2",
          "T29",
          "T30",
1350     "T26",
          "T23",
          "T25",
          "T12",
          "T1",
1355     "T17",
          "T7",
          "T20",
          "T11",
          "T28",
1360     "T10",
          "T32",
          "T18",
          "T24",
          "T14",
1365     "T21",
          "T8",
          "T27",
          "T16",
          "T4",
1370     "T13",
          "T22",
          "T19"
        ],
      "energyUsed": 19489.25,
1375     "lenNotExecTasks": 5,
      "realTime": "1.315µs"
    }
  ]
},
1380 Рыночных" методов": {
  "totalTasks": 32,
  "totalMachine": 8,
  "totalEnergyMachines": 25128,
  "allMachines": [
1385    {
      "id": "M1",
      "equipment": [
          1,
          3,
1390     9
      ],
      "energy": 3141.75,

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    "AssignedTask": null
  },
1395 {
    "id": "M2",
    "equipment": [
      2,
1400     9
    ],
    "energy": 3141.75,
    "AssignedTask": null
  },
1405 {
    "id": "M3",
    "equipment": [
      1,
      7,
1410     14
    ],
    "energy": 3141.75,
    "AssignedTask": null
  },
1415 {
    "id": "M4",
    "equipment": [
      4,
      5,
1420     6,
      15
    ],
    "energy": 3141.75,
    "AssignedTask": null
  },
1425 {
    "id": "M5",
    "equipment": [
      11
    ],
1430     "energy": 3141.75,
    "AssignedTask": null
  },
1435 {
    "id": "M6",
    "equipment": [
      8,
      13
    ],
1440     "energy": 3141.75,
    "AssignedTask": null
  },
  {
    "id": "M7",
    "equipment": [

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1445         10
           ],
           "energy": 3141.75,
           "AssignedTask": null
1450       },
       {
         "id": "M8",
         "equipment": [
           12
         ],
1455       "energy": 3141.75,
       "AssignedTask": null
     }
   ],
   "totalEnergyTasks": 25134,
1460 "allTasks": [
     {
       "id": "T23",
       "required_equipment": 8,
       "energy_cost": 1500,
1465 "CreatedAt": 1759044740
     },
     {
       "id": "T12",
       "required_equipment": 12,
1470 "energy_cost": 10,
       "CreatedAt": 1759044729
     },
     {
       "id": "T30",
1475 "required_equipment": 15,
       "energy_cost": 10,
       "CreatedAt": 1759044747
     },
     {
1480 "id": "T32",
       "required_equipment": 12,
       "energy_cost": 994,
       "CreatedAt": 1759044719
     },
1485 {
       "id": "T20",
       "required_equipment": 5,
       "energy_cost": 10,
       "CreatedAt": 1759044737
1490 },
     {
       "id": "T29",
       "required_equipment": 14,
       "energy_cost": 1500,
1495 "CreatedAt": 1759044746
     },
   ],

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1500      {
            "id": "T25",
            "required_equipment": 10,
            "energy_cost": 1500,
            "CreatedAt": 1759044742
        },
        {
1505      "id": "T9",
            "required_equipment": 9,
            "energy_cost": 1500,
            "CreatedAt": 1759044726
        },
        {
1510      "id": "T21",
            "required_equipment": 6,
            "energy_cost": 1500,
            "CreatedAt": 1759044738
        },
        {
1515      "id": "T28",
            "required_equipment": 13,
            "energy_cost": 10,
            "CreatedAt": 1759044745
        },
1520      },
        {
            "id": "T7",
            "required_equipment": 7,
            "energy_cost": 1500,
1525      "CreatedAt": 1759044724
        },
        {
            "id": "T1",
            "required_equipment": 1,
            "energy_cost": 1500,
1530      "CreatedAt": 1759044718
        },
        {
            "id": "T10",
            "required_equipment": 10,
            "energy_cost": 10,
1535      "CreatedAt": 1759044727
        },
        {
1540      "id": "T8",
            "required_equipment": 8,
            "energy_cost": 10,
            "CreatedAt": 1759044725
        },
        {
1545      "id": "T14",
            "required_equipment": 14,
            "energy_cost": 10,

```



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1550         "CreatedAt": 1759044731
    },
    {
1555         "id": "T18",
        "required_equipment": 3,
        "energy_cost": 10,
        "CreatedAt": 1759044735
    },
    {
1560         "id": "T26",
        "required_equipment": 11,
        "energy_cost": 10,
        "CreatedAt": 1759044743
    },
    {
1565         "id": "T2",
        "required_equipment": 2,
        "energy_cost": 10,
        "CreatedAt": 1759044719
    },
    {
1570         "id": "T4",
        "required_equipment": 4,
        "energy_cost": 10,
        "CreatedAt": 1759044721
    },
    {
1575         "id": "T3",
        "required_equipment": 3,
        "energy_cost": 1500,
        "CreatedAt": 1759044720
    },
    {
1580         "id": "T11",
        "required_equipment": 11,
        "energy_cost": 1500,
        "CreatedAt": 1759044728
    },
    {
1585         "id": "T16",
        "required_equipment": 1,
        "energy_cost": 10,
        "CreatedAt": 1759044733
    },
    {
1590         "id": "T22",
        "required_equipment": 7,
        "energy_cost": 10,
        "CreatedAt": 1759044739
    },
    {
1595         "id": "T17",

```

```

    "required_equipment": 2,
    "energy_cost": 1500,
    "CreatedAt": 1759044734
  },
1605 {
    "id": "T24",
    "required_equipment": 9,
    "energy_cost": 10,
    "CreatedAt": 1759044741
1610 },
  {
    "id": "T19",
    "required_equipment": 4,
    "energy_cost": 1500,
1615 "CreatedAt": 1759044736
  },
  {
    "id": "T27",
    "required_equipment": 12,
1620 "energy_cost": 1500,
    "CreatedAt": 1759044744
  },
  {
    "id": "T13",
    "required_equipment": 13,
1625 "energy_cost": 1500,
    "CreatedAt": 1759044730
  },
  {
    "id": "T31",
    "required_equipment": 7,
1630 "energy_cost": 1490,
    "CreatedAt": 1759044718
  },
  {
    "id": "T15",
    "required_equipment": 15,
1635 "energy_cost": 1500,
    "CreatedAt": 1759044732
  },
1640 {
    "id": "T5",
    "required_equipment": 5,
    "energy_cost": 1500,
1645 "CreatedAt": 1759044722
  },
  {
    "id": "T6",
    "required_equipment": 6,
1650 "energy_cost": 10,
    "CreatedAt": 1759044723
  }

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],
"totalTime": "33.043µs",
1655 "stepMetrics": [
    {
        "step": 1,
        "lenTasksDone": 27,
        "tasksDone": [
1660     "T23",
        "T12",
        "T30",
        "T32",
        "T20",
1665     "T29",
        "T25",
        "T9",
        "T21",
        "T28",
1670     "T7",
        "T1",
        "T10",
        "T8",
        "T14",
1675     "T18",
        "T26",
        "T2",
        "T4",
        "T11",
1680     "T16",
        "T22",
        "T17",
        "T24",
        "T19",
1685     "T27",
        "T13"
    ],
    "energyUsed": 19489.25,
    "lenNotExecTasks": 5,
1690     "realTime": "9.167µs"
},
{
    "step": 2,
    "lenTasksDone": 27,
1695     "tasksDone": [
        "T23",
        "T12",
        "T30",
        "T32",
1700     "T20",
        "T29",
        "T25",
        "T9",
        "T21",

```

```

1705     "T28",
        "T7",
        "T1",
        "T10",
        "T8",
1710     "T14",
        "T18",
        "T26",
        "T2",
        "T4",
1715     "T11",
        "T16",
        "T22",
        "T17",
        "T24",
1720     "T19",
        "T27",
        "T13"
    ],
    "energyUsed": 19489.25,
1725     "lenNotExecTasks": 5,
    "realTime": "5.575µs"
},
{
1730     "step": 3,
    "lenTasksDone": 27,
    "tasksDone": [
        "T23",
        "T12",
        "T30",
1735     "T32",
        "T20",
        "T29",
        "T25",
        "T9",
1740     "T21",
        "T28",
        "T7",
        "T1",
        "T10",
1745     "T8",
        "T14",
        "T18",
        "T26",
        "T2",
1750     "T4",
        "T11",
        "T16",
        "T22",
        "T17",
1755     "T24",
        "T19",

```

```

    "T27",
    "T13"
  ],
  "energyUsed": 19489.25,
  "lenNotExecTasks": 5,
  "realTime": "5.633μs"
},
{
  "step": 4,
  "lenTasksDone": 27,
  "tasksDone": [
    "T23",
    "T12",
    "T30",
    "T32",
    "T20",
    "T29",
    "T25",
    "T9",
    "T21",
    "T28",
    "T7",
    "T1",
    "T10",
    "T8",
    "T14",
    "T18",
    "T26",
    "T2",
    "T4",
    "T11",
    "T16",
    "T22",
    "T17",
    "T24",
    "T19",
    "T27",
    "T13"
  ],
  "energyUsed": 19489.25,
  "lenNotExecTasks": 5,
  "realTime": "4.32μs"
},
{
  "step": 5,
  "lenTasksDone": 27,
  "tasksDone": [
    "T23",
    "T12",
    "T30",
    "T32",
    "T20",

```

```

1810         "T29",
        "T25",
        "T9",
        "T21",
        "T28",
1815        "T7",
        "T1",
        "T10",
        "T8",
        "T14",
        "T18",
1820        "T26",
        "T2",
        "T4",
        "T11",
        "T16",
1825        "T22",
        "T17",
        "T24",
        "T19",
        "T27",
1830        "T13"
    ],
    "energyUsed": 19489.25,
    "lenNotExecTasks": 5,
    "realTime": "4.341µs"
1835 }
    ]
},
Энергосберегающая": {
    "totalTasks": 32,
1840    "totalMachine": 8,
    "totalEnergyMachines": 25128,
    "allMachines": [
        {
1845            "id": "M1",
            "equipment": [
                1,
                3,
                9
            ],
1850            "energy": 3141.75,
            "AssignedTask": null
        },
        {
1855            "id": "M2",
            "equipment": [
                2,
                9
            ],
            "energy": 3141.75,
1860            "AssignedTask": null
        }
    ]
}

```

	},
	{
	"id": "M3",
1865	"equipment": [
	1,
	7,
	14
	],
	"energy": 3141.75,
1870	"AssignedTask": null
	},
	{
	"id": "M4",
1875	"equipment": [
	4,
	5,
	6,
	15
	],
1880	"energy": 3141.75,
	"AssignedTask": null
	},
	{
1885	"id": "M5",
	"equipment": [
	11
	],
	"energy": 3141.75,
1890	"AssignedTask": null
	},
	{
	"id": "M6",
	"equipment": [
1895	8,
	13
	],
	"energy": 3141.75,
	"AssignedTask": null
	},
1900	{
	"id": "M7",
	"equipment": [
	10
	],
1905	"energy": 3141.75,
	"AssignedTask": null
	},
	{
1910	"id": "M8",
	"equipment": [
	12
	],

```

    "energy": 3141.75,
    "AssignedTask": null
1915  }
],
"totalEnergyTasks": 25134,
"allTasks": [
1920  {
    "id": "T23",
    "required_equipment": 8,
    "energy_cost": 1500,
    "CreatedAt": 1759044740
1925  },
  {
    "id": "T12",
    "required_equipment": 12,
    "energy_cost": 10,
    "CreatedAt": 1759044729
1930  },
  {
    "id": "T30",
    "required_equipment": 15,
    "energy_cost": 10,
1935    "CreatedAt": 1759044747
  },
  {
    "id": "T32",
    "required_equipment": 12,
1940    "energy_cost": 994,
    "CreatedAt": 1759044719
  },
  {
    "id": "T20",
    "required_equipment": 5,
1945    "energy_cost": 10,
    "CreatedAt": 1759044737
  },
  {
1950    "id": "T29",
    "required_equipment": 14,
    "energy_cost": 1500,
    "CreatedAt": 1759044746
1955  },
  {
    "id": "T25",
    "required_equipment": 10,
    "energy_cost": 1500,
    "CreatedAt": 1759044742
1960  },
  {
    "id": "T9",
    "required_equipment": 9,
    "energy_cost": 1500,

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1965	"CreatedAt": 1759044726
	},
	{
	"id": "T21",
	"required_equipment": 6,
1970	"energy_cost": 1500,
	"CreatedAt": 1759044738
	},
	{
	"id": "T28",
1975	"required_equipment": 13,
	"energy_cost": 10,
	"CreatedAt": 1759044745
	},
	{
1980	"id": "T7",
	"required_equipment": 7,
	"energy_cost": 1500,
	"CreatedAt": 1759044724
	},
1985	{
	"id": "T1",
	"required_equipment": 1,
	"energy_cost": 1500,
	"CreatedAt": 1759044718
1990	},
	{
	"id": "T10",
	"required_equipment": 10,
	"energy_cost": 10,
1995	"CreatedAt": 1759044727
	},
	{
	"id": "T8",
	"required_equipment": 8,
2000	"energy_cost": 10,
	"CreatedAt": 1759044725
	},
	{
	"id": "T14",
2005	"required_equipment": 14,
	"energy_cost": 10,
	"CreatedAt": 1759044731
	},
	{
2010	"id": "T18",
	"required_equipment": 3,
	"energy_cost": 10,
	"CreatedAt": 1759044735
	},
2015	{
	"id": "T26",

	<pre> "required_equipment": 11, "energy_cost": 10, "CreatedAt": 1759044743 }, { "id": "T2", "required_equipment": 2, "energy_cost": 10, "CreatedAt": 1759044719 }, { "id": "T4", "required_equipment": 4, "energy_cost": 10, "CreatedAt": 1759044721 }, { "id": "T3", "required_equipment": 3, "energy_cost": 1500, "CreatedAt": 1759044720 }, { "id": "T11", "required_equipment": 11, "energy_cost": 1500, "CreatedAt": 1759044728 }, { "id": "T16", "required_equipment": 1, "energy_cost": 10, "CreatedAt": 1759044733 }, { "id": "T22", "required_equipment": 7, "energy_cost": 10, "CreatedAt": 1759044739 }, { "id": "T17", "required_equipment": 2, "energy_cost": 1500, "CreatedAt": 1759044734 }, { "id": "T24", "required_equipment": 9, "energy_cost": 10, "CreatedAt": 1759044741 }, }, </pre>
2020	
2025	
2030	
2035	
2040	
2045	
2050	
2055	
2060	
2065	

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2070     {
        "id": "T19",
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        "energy_cost": 1500,
        "CreatedAt": 1759044736
2075     },
    {
        "id": "T27",
        "required_equipment": 12,
        "energy_cost": 1500,
        "CreatedAt": 1759044744
2080     },
    {
        "id": "T13",
        "required_equipment": 13,
        "energy_cost": 1500,
        "CreatedAt": 1759044730
2085     },
    {
        "id": "T31",
        "required_equipment": 7,
        "energy_cost": 1490,
        "CreatedAt": 1759044718
2090     },
    {
        "id": "T15",
        "required_equipment": 15,
        "energy_cost": 1500,
        "CreatedAt": 1759044732
2095     },
    {
        "id": "T5",
        "required_equipment": 5,
        "energy_cost": 1500,
        "CreatedAt": 1759044722
2100     },
    {
        "id": "T6",
        "required_equipment": 6,
        "energy_cost": 10,
        "CreatedAt": 1759044723
2105     },
    }
2110 ],
    "totalTime": "41.369µs",
    "stepMetrics": [
2115     {
        "step": 1,
        "lenTasksDone": 28,
        "tasksDone": [
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            "T2",
            "T14",
2120

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2125     "T30",
        "T26",
        "T28",
        "T10",
        "T12",
        "T16",
        "T24",
        "T22",
        "T20",
2130     "T11",
        "T8",
        "T25",
        "T32",
        "T9",
2135     "T17",
        "T31",
        "T4",
        "T23",
        "T27",
2140     "T1",
        "T29",
        "T6",
        "T13",
        "T21",
2145     "T19"
    ],
    "energyUsed": 19489.25,
    "lenNotExecTasks": 4,
    "realTime": "12.535µs"
2150 },
    {
        "step": 2,
        "lenTasksDone": 28,
        "tasksDone": [
2155     "T18",
        "T2",
        "T14",
        "T30",
        "T26",
2160     "T28",
        "T10",
        "T12",
        "T16",
        "T24",
2165     "T22",
        "T20",
        "T11",
        "T8",
        "T25",
2170     "T32",
        "T9",
        "T17",

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2175         "T31",
            "T4",
            "T23",
            "T27",
            "T1",
            "T29",
            "T6",
2180         "T13",
            "T21",
            "T19"
        ],
        "energyUsed": 19489.25,
2185         "lenNotExecTasks": 4,
        "realTime": "6.934µs"
    },
    {
2190         "step": 3,
        "lenTasksDone": 28,
        "tasksDone": [
            "T18",
            "T2",
            "T14",
2195         "T30",
            "T26",
            "T28",
            "T10",
            "T12",
            "T16",
2200         "T24",
            "T22",
            "T20",
            "T11",
            "T8",
2205         "T25",
            "T32",
            "T9",
            "T17",
            "T31",
2210         "T4",
            "T23",
            "T27",
            "T1",
            "T29",
2215         "T6",
            "T13",
            "T21",
            "T19"
        ],
2220         "energyUsed": 19489.25,
        "lenNotExecTasks": 4,
        "realTime": "5.632µs"
    },

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2225 {
    "step": 4,
    "lenTasksDone": 28,
    "tasksDone": [
2230     "T18",
        "T2",
        "T14",
        "T30",
        "T26",
        "T28",
2235     "T10",
        "T12",
        "T16",
        "T24",
        "T22",
2240     "T20",
        "T11",
        "T8",
        "T25",
        "T32",
2245     "T9",
        "T17",
        "T31",
        "T4",
        "T23",
2250     "T27",
        "T1",
        "T29",
        "T6",
        "T13",
2255     "T21",
        "T19"
    ],
    "energyUsed": 19489.25,
    "lenNotExecTasks": 4,
2260     "realTime": "4.859µs"
},
{
    "step": 5,
    "lenTasksDone": 28,
2265     "tasksDone": [
        "T18",
        "T2",
        "T14",
        "T30",
2270     "T26",
        "T28",
        "T10",
        "T12",
        "T16",
2275     "T24",
        "T22",

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2280         "T20",
            "T11",
            "T8",
            "T25",
            "T32",
            "T9",
            "T17",
            "T31",
2285         "T4",
            "T23",
            "T27",
            "T1",
            "T29",
2290         "T6",
            "T13",
            "T21",
            "T19"
        ],
2295         "energyUsed": 19489.25,
        "lenNotExecTasks": 4,
        "realTime": "5.69μs"
    }
}
2300 },
    "max": {
        "RL (Q-learning)": {
2305         "totalTasks": 32,
            "totalMachine": 8,
            "totalEnergyMachines": 50264,
            "allMachines": [
                {
2310                 "id": "M1",
                    "equipment": [
                        1,
                        3,
                        9
                    ],
2315                 "energy": 6283.5,
                    "AssignedTask": null
                },
                {
2320                 "id": "M2",
                    "equipment": [
                        2,
                        9
                    ],
                    "energy": 6283.5,
2325                 "AssignedTask": null
                },
                {
                    "id": "M3",

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2330         "equipment": [
            1,
            7,
            14
        ],
        "energy": 6283.5,
2335     "AssignedTask": null
    },
    {
        "id": "M4",
        "equipment": [
2340            4,
            5,
            6,
            15
        ],
        "energy": 6283.5,
2345     "AssignedTask": null
    },
    {
        "id": "M5",
2350     "equipment": [
            11
        ],
        "energy": 6283.5,
        "AssignedTask": null
2355 },
    {
        "id": "M6",
        "equipment": [
2360            8,
            13
        ],
        "energy": 6283.5,
        "AssignedTask": null
2365 },
    {
        "id": "M7",
        "equipment": [
            10
        ],
        "energy": 6283.5,
2370     "AssignedTask": null
    },
    {
        "id": "M8",
2375     "equipment": [
            12
        ],
        "energy": 6283.5,
        "AssignedTask": null
2380 }

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],
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"allTasks": [
2385     {
        "id": "T23",
        "required_equipment": 8,
        "energy_cost": 1500,
        "CreatedAt": 1759044740
2390     },
        {
            "id": "T12",
            "required_equipment": 12,
            "energy_cost": 10,
            "CreatedAt": 1759044729
2395     },
        {
            "id": "T30",
            "required_equipment": 15,
            "energy_cost": 10,
2400            "CreatedAt": 1759044747
        },
        {
            "id": "T32",
            "required_equipment": 12,
2405            "energy_cost": 994,
            "CreatedAt": 1759044719
        },
        {
            "id": "T20",
2410            "required_equipment": 5,
            "energy_cost": 10,
            "CreatedAt": 1759044737
        },
        {
2415            "id": "T29",
            "required_equipment": 14,
            "energy_cost": 1500,
            "CreatedAt": 1759044746
        },
2420     },
        {
            "id": "T25",
            "required_equipment": 10,
            "energy_cost": 1500,
            "CreatedAt": 1759044742
2425     },
        {
            "id": "T9",
            "required_equipment": 9,
            "energy_cost": 1500,
2430            "CreatedAt": 1759044726
        },
        {

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2435     "id": "T21",
        "required_equipment": 6,
        "energy_cost": 1500,
        "CreatedAt": 1759044738
    },
    {
2440     "id": "T28",
        "required_equipment": 13,
        "energy_cost": 10,
        "CreatedAt": 1759044745
    },
    {
2445     "id": "T7",
        "required_equipment": 7,
        "energy_cost": 1500,
        "CreatedAt": 1759044724
    },
    {
2450     "id": "T1",
        "required_equipment": 1,
        "energy_cost": 1500,
        "CreatedAt": 1759044718
2455 },
    {
        "id": "T10",
        "required_equipment": 10,
        "energy_cost": 10,
2460     "CreatedAt": 1759044727
    },
    {
        "id": "T8",
        "required_equipment": 8,
2465     "energy_cost": 10,
        "CreatedAt": 1759044725
    },
    {
        "id": "T14",
2470     "required_equipment": 14,
        "energy_cost": 10,
        "CreatedAt": 1759044731
    },
    {
2475     "id": "T18",
        "required_equipment": 3,
        "energy_cost": 10,
        "CreatedAt": 1759044735
    },
    {
2480     "id": "T26",
        "required_equipment": 11,
        "energy_cost": 10,
        "CreatedAt": 1759044743

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```

2485     },
        {
            "id": "T2",
            "required_equipment": 2,
            "energy_cost": 10,
2490     "CreatedAt": 1759044719
        },
        {
            "id": "T4",
            "required_equipment": 4,
2495     "energy_cost": 10,
            "CreatedAt": 1759044721
        },
        {
            "id": "T3",
2500     "required_equipment": 3,
            "energy_cost": 1500,
            "CreatedAt": 1759044720
        },
        {
2505     "id": "T11",
            "required_equipment": 11,
            "energy_cost": 1500,
            "CreatedAt": 1759044728
        },
2510     {
            "id": "T16",
            "required_equipment": 1,
            "energy_cost": 10,
            "CreatedAt": 1759044733
2515     },
        {
            "id": "T22",
            "required_equipment": 7,
            "energy_cost": 10,
2520     "CreatedAt": 1759044739
        },
        {
            "id": "T17",
            "required_equipment": 2,
2525     "energy_cost": 1500,
            "CreatedAt": 1759044734
        },
        {
            "id": "T24",
2530     "required_equipment": 9,
            "energy_cost": 10,
            "CreatedAt": 1759044741
        },
        {
2535     "id": "T19",
            "required_equipment": 4,

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    "energy_cost": 1500,
    "CreatedAt": 1759044736
  },
2540  {
    "id": "T27",
    "required_equipment": 12,
    "energy_cost": 1500,
    "CreatedAt": 1759044744
2545  },
  {
    "id": "T13",
    "required_equipment": 13,
    "energy_cost": 1500,
2550  "CreatedAt": 1759044730
  },
  {
    "id": "T31",
    "required_equipment": 7,
2555  "energy_cost": 1490,
    "CreatedAt": 1759044718
  },
  {
    "id": "T15",
2560  "required_equipment": 15,
    "energy_cost": 1500,
    "CreatedAt": 1759044732
  },
  {
2565  "id": "T5",
    "required_equipment": 5,
    "energy_cost": 1500,
    "CreatedAt": 1759044722
  },
2570  {
    "id": "T6",
    "required_equipment": 6,
    "energy_cost": 10,
    "CreatedAt": 1759044723
2575  }
  ],
  "totalTime": "156.989µs",
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2580  {
    "step": 1,
    "lenTasksDone": 32,
    "tasksDone": [
2585  "T16",
    "T2",
    "T14",
    "T20",
    "T26",
    "T8",

```

```

2590     "T10",
        "T12",
        "T1",
        "T17",
        "T29",
        "T5",
2595     "T11",
        "T23",
        "T25",
        "T32",
        "T18",
2600     "T24",
        "T22",
        "T30",
        "T28",
        "T27",
2605     "T3",
        "T9",
        "T31",
        "T15",
        "T13",
2610     "T7",
        "T6",
        "T21",
        "T4",
        "T19"
2615 ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,
        "realTime": "34.845µs"
    },
2620 {
        "step": 2,
        "lenTasksDone": 32,
        "tasksDone": [
2625     "T16",
        "T2",
        "T14",
        "T20",
        "T26",
        "T8",
2630     "T10",
        "T12",
        "T1",
        "T17",
        "T22",
2635     "T5",
        "T11",
        "T23",
        "T25",
        "T32",
2640     "T18",

```

```

2645     "T24",
        "T29",
        "T30",
        "T28",
        "T27",
        "T3",
        "T9",
        "T31",
        "T15",
2650     "T13",
        "T7",
        "T6",
        "T21",
        "T4",
2655     "T19"
    ],
    "energyUsed": 25134,
    "lenNotExecTasks": 0,
    "realTime": "33.032μs"
2660 },
    {
        "step": 3,
        "lenTasksDone": 32,
        "tasksDone": [
2665     "T16",
        "T2",
        "T14",
        "T20",
        "T26",
2670     "T8",
        "T10",
        "T12",
        "T1",
        "T17",
2675     "T29",
        "T5",
        "T11",
        "T23",
        "T25",
2680     "T32",
        "T18",
        "T24",
        "T22",
        "T30",
2685     "T28",
        "T27",
        "T3",
        "T9",
        "T31",
2690     "T15",
        "T13",
        "T7",

```

```

2695         "T6",
            "T21",
            "T4",
            "T19"
        ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,
2700     "realTime": "26.235µs"
    },
    {
        "step": 4,
        "lenTasksDone": 32,
2705     "tasksDone": [
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            "T2",
            "T1",
            "T20",
2710     "T26",
            "T8",
            "T10",
            "T12",
            "T18",
2715     "T17",
            "T22",
            "T5",
            "T11",
            "T23",
2720     "T25",
            "T32",
            "T24",
            "T9",
            "T31",
2725     "T30",
            "T28",
            "T27",
            "T3",
            "T7",
2730     "T15",
            "T13",
            "T14",
            "T6",
            "T29",
2735     "T21",
            "T4",
            "T19"
        ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,
2740     "realTime": "30.291µs"
    },
    {
        "step": 5,

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2745         "lenTasksDone": 32,
        "tasksDone": [
            "T24",
            "T2",
            "T22",
2750         "T20",
            "T26",
            "T8",
            "T10",
            "T12",
2755         "T16",
            "T17",
            "T31",
            "T5",
            "T11",
2760         "T23",
            "T25",
            "T32",
            "T1",
            "T9",
2765         "T14",
            "T30",
            "T28",
            "T27",
            "T18",
2770         "T7",
            "T15",
            "T13",
            "T3",
            "T29",
2775         "T6",
            "T21",
            "T4",
            "T19"
        ],
2780         "energyUsed": 25134,
        "lenNotExecTasks": 0,
        "realTime": "28.626µs"
    }
    ],
2785 },
    Минимизация" Вр3": {
        "totalTasks": 32,
        "totalMachine": 8,
        "totalEnergyMachines": 50264,
2790         "allMachines": [
            {
                "id": "M1",
                "equipment": [
2795                 1,
                 3,
                 9
            ]
        ]
    }
}

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    ],
    "energy": 6283.5,
    "AssignedTask": null
2800 },
    {
        "id": "M2",
        "equipment": [
2805     2,
        9
        ],
        "energy": 6283.5,
        "AssignedTask": null
    },
2810 {
    "id": "M3",
    "equipment": [
        1,
2815     7,
        14
    ],
    "energy": 6283.5,
    "AssignedTask": null
},
2820 {
    "id": "M4",
    "equipment": [
        4,
2825     5,
        6,
        15
    ],
    "energy": 6283.5,
    "AssignedTask": null
2830 },
    {
        "id": "M5",
        "equipment": [
2835     11
        ],
        "energy": 6283.5,
        "AssignedTask": null
    },
2840 {
    "id": "M6",
    "equipment": [
        8,
2845     13
        ],
        "energy": 6283.5,
        "AssignedTask": null
    },
    {

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```

2850         "id": "M7",
           "equipment": [
             10
           ],
           "energy": 6283.5,
           "AssignedTask": null
2855     },
     {
       "id": "M8",
       "equipment": [
2860         12
       ],
       "energy": 6283.5,
       "AssignedTask": null
     }
   ],
2865   "totalEnergyTasks": 25134,
   "allTasks": [
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       "id": "T23",
       "required_equipment": 8,
2870       "energy_cost": 1500,
       "CreatedAt": 1759044740
     },
     {
       "id": "T12",
2875       "required_equipment": 12,
       "energy_cost": 10,
       "CreatedAt": 1759044729
     },
     {
2880       "id": "T30",
       "required_equipment": 15,
       "energy_cost": 10,
       "CreatedAt": 1759044747
     },
2885     {
       "id": "T32",
       "required_equipment": 12,
       "energy_cost": 994,
       "CreatedAt": 1759044719
2890     },
     {
       "id": "T20",
       "required_equipment": 5,
       "energy_cost": 10,
2895       "CreatedAt": 1759044737
     },
     {
       "id": "T29",
       "required_equipment": 14,
2900       "energy_cost": 1500,

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    "CreatedAt": 1759044746
  },
  {
    "id": "T25",
    "required_equipment": 10,
    "energy_cost": 1500,
    "CreatedAt": 1759044742
  },
  {
    "id": "T9",
    "required_equipment": 9,
    "energy_cost": 1500,
    "CreatedAt": 1759044726
  },
  {
    "id": "T21",
    "required_equipment": 6,
    "energy_cost": 1500,
    "CreatedAt": 1759044738
  },
  {
    "id": "T28",
    "required_equipment": 13,
    "energy_cost": 10,
    "CreatedAt": 1759044745
  },
  {
    "id": "T7",
    "required_equipment": 7,
    "energy_cost": 1500,
    "CreatedAt": 1759044724
  },
  {
    "id": "T1",
    "required_equipment": 1,
    "energy_cost": 1500,
    "CreatedAt": 1759044718
  },
  {
    "id": "T10",
    "required_equipment": 10,
    "energy_cost": 10,
    "CreatedAt": 1759044727
  },
  {
    "id": "T8",
    "required_equipment": 8,
    "energy_cost": 10,
    "CreatedAt": 1759044725
  },
  {
    "id": "T14",

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2955     "required_equipment": 14,
        "energy_cost": 10,
        "CreatedAt": 1759044731
    },
    {
2960     "id": "T18",
        "required_equipment": 3,
        "energy_cost": 10,
        "CreatedAt": 1759044735
    },
    {
2965     "id": "T26",
        "required_equipment": 11,
        "energy_cost": 10,
        "CreatedAt": 1759044743
    },
    {
2970     "id": "T2",
        "required_equipment": 2,
        "energy_cost": 10,
        "CreatedAt": 1759044719
    },
    {
2975     "id": "T4",
        "required_equipment": 4,
        "energy_cost": 10,
        "CreatedAt": 1759044721
    },
    {
2980     "id": "T3",
        "required_equipment": 3,
        "energy_cost": 1500,
2985     "CreatedAt": 1759044720
    },
    {
        "id": "T11",
2990     "required_equipment": 11,
        "energy_cost": 1500,
        "CreatedAt": 1759044728
    },
    {
2995     "id": "T16",
        "required_equipment": 1,
        "energy_cost": 10,
        "CreatedAt": 1759044733
    },
    {
3000     "id": "T22",
        "required_equipment": 7,
        "energy_cost": 10,
        "CreatedAt": 1759044739
    },
    },

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3005 {
      "id": "T17",
      "required_equipment": 2,
      "energy_cost": 1500,
      "CreatedAt": 1759044734
3010 },
      {
      "id": "T24",
      "required_equipment": 9,
      "energy_cost": 10,
3015 "CreatedAt": 1759044741
      },
      {
      "id": "T19",
      "required_equipment": 4,
3020 "energy_cost": 1500,
      "CreatedAt": 1759044736
      },
      {
      "id": "T27",
3025 "required_equipment": 12,
      "energy_cost": 1500,
      "CreatedAt": 1759044744
      },
      {
3030 "id": "T13",
      "required_equipment": 13,
      "energy_cost": 1500,
      "CreatedAt": 1759044730
      },
3035 {
      "id": "T31",
      "required_equipment": 7,
      "energy_cost": 1490,
      "CreatedAt": 1759044718
3040 },
      {
      "id": "T15",
      "required_equipment": 15,
      "energy_cost": 1500,
3045 "CreatedAt": 1759044732
      },
      {
      "id": "T5",
3050 "required_equipment": 5,
      "energy_cost": 1500,
      "CreatedAt": 1759044722
      },
      {
3055 "id": "T6",
      "required_equipment": 6,
      "energy_cost": 10,

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    "CreatedAt": 1759044723
  }
],
3060 "totalTime": "85.487µs",
"stepMetrics": [
  {
    "step": 1,
    "lenTasksDone": 32,
3065 "tasksDone": [
      "T23",
      "T12",
      "T30",
      "T29",
3070 "T25",
      "T9",
      "T26",
      "T32",
      "T20",
3075 "T28",
      "T7",
      "T1",
      "T10",
      "T2",
3080 "T11",
      "T21",
      "T8",
      "T14",
      "T18",
3085 "T17",
      "T27",
      "T4",
      "T3",
      "T16",
3090 "T24",
      "T13",
      "T22",
      "T19",
      "T31",
3095 "T15",
      "T5",
      "T6"
    ],
    "energyUsed": 25134,
3100 "lenNotExecTasks": 0,
    "realTime": "18.91µs"
  },
  {
    "step": 2,
3105 "lenTasksDone": 32,
    "tasksDone": [
      "T23",
      "T12",

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3110      "T30",
      "T29",
      "T25",
      "T9",
      "T26",
3115      "T32",
      "T20",
      "T28",
      "T7",
      "T1",
      "T10",
3120      "T2",
      "T11",
      "T21",
      "T8",
      "T14",
3125      "T18",
      "T17",
      "T27",
      "T4",
      "T3",
3130      "T16",
      "T24",
      "T13",
      "T22",
      "T19",
3135      "T31",
      "T15",
      "T5",
      "T6"
    ],
3140    "energyUsed": 25134,
    "lenNotExecTasks": 0,
    "realTime": "17.628µs"
  },
  {
3145    "step": 3,
    "lenTasksDone": 32,
    "tasksDone": [
      "T23",
      "T12",
3150      "T30",
      "T29",
      "T25",
      "T9",
      "T26",
3155      "T32",
      "T20",
      "T28",
      "T7",
      "T1",
3160      "T10",

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3165     "T2",
        "T11",
        "T21",
        "T8",
        "T14",
        "T18",
        "T17",
        "T27",
        "T4",
3170     "T3",
        "T16",
        "T24",
        "T13",
        "T22",
        "T19",
        "T31",
        "T15",
        "T5",
        "T6"
3180 ],
    "energyUsed": 25134,
    "lenNotExecTasks": 0,
    "realTime": "18.697µs"
3185 },
    {
        "step": 4,
        "lenTasksDone": 32,
        "tasksDone": [
3190     "T23",
        "T12",
        "T30",
        "T29",
        "T25",
        "T9",
        "T26",
        "T32",
        "T20",
        "T28",
        "T7",
3200     "T1",
        "T10",
        "T2",
        "T11",
        "T21",
        "T8",
        "T14",
        "T18",
        "T17",
        "T27",
        "T4",
3210     "T3",
        "T16",

```



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3215         "T24",
            "T13",
            "T22",
            "T19",
            "T31",
            "T15",
            "T5",
3220         "T6"
        ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,
        "realTime": "12.329µs"
3225     },
    {
        "step": 5,
        "lenTasksDone": 32,
        "tasksDone": [
3230         "T23",
            "T12",
            "T30",
            "T29",
            "T25",
3235         "T9",
            "T26",
            "T32",
            "T20",
            "T28",
3240         "T7",
            "T1",
            "T10",
            "T2",
            "T11",
3245         "T21",
            "T8",
            "T14",
            "T18",
            "T17",
3250         "T27",
            "T4",
            "T3",
            "T16",
            "T24",
3255         "T13",
            "T22",
            "T19",
            "T31",
            "T15",
3260         "T5",
            "T6"
        ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,

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3265         "realTime": "13.46µs"
        }
    ]
},
Прямого" перебора": {
3270     "totalTasks": 32,
    "totalMachine": 8,
    "totalEnergyMachines": 50264,
    "allMachines": [
3275        {
            "id": "M1",
            "equipment": [
                1,
                3,
                9
3280            ],
            "energy": 6283.5,
            "AssignedTask": null
        },
        {
3285            "id": "M2",
            "equipment": [
                2,
                9
            ],
            "energy": 6283.5,
            "AssignedTask": null
3290        },
        {
            "id": "M3",
            "equipment": [
3295                1,
                7,
                14
            ],
            "energy": 6283.5,
            "AssignedTask": null
3300        },
        {
            "id": "M4",
            "equipment": [
3305                4,
                5,
                6,
                15
3310            ],
            "energy": 6283.5,
            "AssignedTask": null
        },
        {
3315            "id": "M5",
            "equipment": [

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```

        11
    ],
    "energy": 6283.5,
    "AssignedTask": null
},
{
    "id": "M6",
    "equipment": [
        8,
        13
    ],
    "energy": 6283.5,
    "AssignedTask": null
},
{
    "id": "M7",
    "equipment": [
        10
    ],
    "energy": 6283.5,
    "AssignedTask": null
},
{
    "id": "M8",
    "equipment": [
        12
    ],
    "energy": 6283.5,
    "AssignedTask": null
}
],
"totalEnergyTasks": 25134,
"allTasks": [
    {
        "id": "T23",
        "required_equipment": 8,
        "energy_cost": 1500,
        "CreatedAt": 1759044740
    },
    {
        "id": "T12",
        "required_equipment": 12,
        "energy_cost": 10,
        "CreatedAt": 1759044729
    },
    {
        "id": "T30",
        "required_equipment": 15,
        "energy_cost": 10,
        "CreatedAt": 1759044747
    },
    {

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3370         "id": "T32",
           "required_equipment": 12,
           "energy_cost": 994,
           "CreatedAt": 1759044719
3375     },
           {
           "id": "T20",
           "required_equipment": 5,
           "energy_cost": 10,
           "CreatedAt": 1759044737
3380     },
           {
           "id": "T29",
           "required_equipment": 14,
           "energy_cost": 1500,
           "CreatedAt": 1759044746
3385     },
           {
           "id": "T25",
           "required_equipment": 10,
           "energy_cost": 1500,
           "CreatedAt": 1759044742
3390     },
           {
           "id": "T9",
           "required_equipment": 9,
           "energy_cost": 1500,
           "CreatedAt": 1759044726
3395     },
           {
           "id": "T21",
           "required_equipment": 6,
           "energy_cost": 1500,
           "CreatedAt": 1759044738
3400     },
           {
           "id": "T28",
           "required_equipment": 13,
           "energy_cost": 10,
           "CreatedAt": 1759044745
3405     },
           {
           "id": "T7",
           "required_equipment": 7,
           "energy_cost": 1500,
           "CreatedAt": 1759044724
3410     },
           {
           "id": "T1",
           "required_equipment": 1,
           "energy_cost": 1500,
           "CreatedAt": 1759044718
3415
3420

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```

},
{
  "id": "T10",
  "required_equipment": 10,
  "energy_cost": 10,
  "CreatedAt": 1759044727
},
{
  "id": "T8",
  "required_equipment": 8,
  "energy_cost": 10,
  "CreatedAt": 1759044725
},
{
  "id": "T14",
  "required_equipment": 14,
  "energy_cost": 10,
  "CreatedAt": 1759044731
},
{
  "id": "T18",
  "required_equipment": 3,
  "energy_cost": 10,
  "CreatedAt": 1759044735
},
{
  "id": "T26",
  "required_equipment": 11,
  "energy_cost": 10,
  "CreatedAt": 1759044743
},
{
  "id": "T2",
  "required_equipment": 2,
  "energy_cost": 10,
  "CreatedAt": 1759044719
},
{
  "id": "T4",
  "required_equipment": 4,
  "energy_cost": 10,
  "CreatedAt": 1759044721
},
{
  "id": "T3",
  "required_equipment": 3,
  "energy_cost": 1500,
  "CreatedAt": 1759044720
},
{
  "id": "T11",
  "required_equipment": 11,

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    "energy_cost": 1500,
    "CreatedAt": 1759044728
3475 },
    {
        "id": "T16",
        "required_equipment": 1,
        "energy_cost": 10,
3480 "CreatedAt": 1759044733
    },
    {
        "id": "T22",
        "required_equipment": 7,
3485 "energy_cost": 10,
        "CreatedAt": 1759044739
    },
    {
        "id": "T17",
        "required_equipment": 2,
3490 "energy_cost": 1500,
        "CreatedAt": 1759044734
    },
    {
        "id": "T24",
        "required_equipment": 9,
3495 "energy_cost": 10,
        "CreatedAt": 1759044741
    },
    {
        "id": "T19",
        "required_equipment": 4,
3500 "energy_cost": 1500,
        "CreatedAt": 1759044736
    },
    {
        "id": "T27",
        "required_equipment": 12,
3505 "energy_cost": 1500,
        "CreatedAt": 1759044744
    },
    {
        "id": "T13",
        "required_equipment": 13,
3510 "energy_cost": 1500,
        "CreatedAt": 1759044730
    },
    {
        "id": "T31",
        "required_equipment": 7,
3515 "energy_cost": 1490,
        "CreatedAt": 1759044718
    },
    {

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3525         "id": "T15",
           "required_equipment": 15,
           "energy_cost": 1500,
           "CreatedAt": 1759044732
3530     },
           {
           "id": "T5",
           "required_equipment": 5,
           "energy_cost": 1500,
           "CreatedAt": 1759044722
3535     },
           {
           "id": "T6",
           "required_equipment": 6,
           "energy_cost": 10,
3540     "CreatedAt": 1759044723
           }
       ],
       "totalTime": "18.605µs",
       "stepMetrics": [
3545     {
           "step": 1,
           "lenTasksDone": 32,
           "tasksDone": [
3550             "T9",
             "T2",
             "T29",
             "T30",
             "T26",
             "T23",
3555             "T25",
             "T12",
             "T1",
             "T17",
             "T7",
3560             "T20",
             "T11",
             "T28",
             "T10",
             "T32",
3565             "T18",
             "T24",
             "T14",
             "T21",
             "T8",
3570             "T27",
             "T3",
             "T16",
             "T4",
             "T13",
3575             "T22",
             "T19",

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3580         "T31",
            "T15",
            "T5",
            "T6"
        ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,
        "realTime": "3.863μs"
3585     },
    {
        "step": 2,
        "lenTasksDone": 32,
        "tasksDone": [
3590            "T9",
            "T2",
            "T29",
            "T30",
            "T26",
3595            "T23",
            "T25",
            "T12",
            "T1",
            "T17",
3600            "T7",
            "T20",
            "T11",
            "T28",
            "T10",
3605            "T32",
            "T18",
            "T24",
            "T14",
            "T21",
3610            "T8",
            "T27",
            "T3",
            "T16",
            "T4",
3615            "T13",
            "T22",
            "T19",
            "T31",
            "T15",
3620            "T5",
            "T6"
        ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,
        "realTime": "2.816μs"
3625     },
    {
        "step": 3,

```



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3630      "lenTasksDone": 32,
      "tasksDone": [
        "T9",
        "T2",
        "T29",
        "T30",
3635      "T26",
        "T23",
        "T25",
        "T12",
        "T1",
3640      "T17",
        "T7",
        "T20",
        "T11",
        "T28",
3645      "T10",
        "T32",
        "T18",
        "T24",
        "T14",
3650      "T21",
        "T8",
        "T27",
        "T3",
        "T16",
3655      "T4",
        "T13",
        "T22",
        "T19",
        "T31",
3660      "T15",
        "T5",
        "T6"
      ],
      "energyUsed": 25134,
3665      "lenNotExecTasks": 0,
      "realTime": "2μs"
    },
    {
3670      "step": 4,
      "lenTasksDone": 32,
      "tasksDone": [
        "T9",
        "T2",
        "T29",
3675      "T30",
        "T26",
        "T23",
        "T25",
        "T12",
3680      "T1",

```

```

3685     "T17",
        "T7",
        "T20",
        "T11",
        "T28",
        "T10",
        "T32",
        "T18",
        "T24",
3690     "T14",
        "T21",
        "T8",
        "T27",
        "T3",
3695     "T16",
        "T4",
        "T13",
        "T22",
        "T19",
3700     "T31",
        "T15",
        "T5",
        "T6"
    ],
3705     "energyUsed": 25134,
        "lenNotExecTasks": 0,
        "realTime": "1.854µs"
    },
    {
3710     "step": 5,
        "lenTasksDone": 32,
        "tasksDone": [
            "T9",
            "T2",
3715     "T29",
            "T30",
            "T26",
            "T23",
            "T25",
3720     "T12",
            "T1",
            "T17",
            "T7",
            "T20",
3725     "T11",
            "T28",
            "T10",
            "T32",
            "T18",
3730     "T24",
            "T14",
            "T21",

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3735         "T8",
            "T27",
            "T3",
            "T16",
            "T4",
            "T13",
            "T22",
3740         "T19",
            "T31",
            "T15",
            "T5",
            "T6"
3745     ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,
        "realTime": "1.646µs"
    }
3750 ]
},
"Рыночных" методов": {
    "totalTasks": 32,
    "totalMachine": 8,
3755    "totalEnergyMachines": 50264,
    "allMachines": [
        {
            "id": "M1",
            "equipment": [
3760                1,
                3,
                9
            ],
            "energy": 6283.5,
3765            "AssignedTask": null
        },
        {
            "id": "M2",
            "equipment": [
3770                2,
                9
            ],
            "energy": 6283.5,
            "AssignedTask": null
3775        },
        {
            "id": "M3",
            "equipment": [
3780                1,
                7,
                14
            ],
            "energy": 6283.5,
            "AssignedTask": null

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3785     },
        {
            "id": "M4",
            "equipment": [
3790                4,
                5,
                6,
                15
            ],
            "energy": 6283.5,
3795            "AssignedTask": null
        },
        {
            "id": "M5",
            "equipment": [
3800                11
            ],
            "energy": 6283.5,
            "AssignedTask": null
        },
3805     {
        "id": "M6",
        "equipment": [
            8,
            13
3810        ],
        "energy": 6283.5,
        "AssignedTask": null
    },
    {
3815        "id": "M7",
        "equipment": [
            10
        ],
        "energy": 6283.5,
3820        "AssignedTask": null
    },
    {
        "id": "M8",
        "equipment": [
3825            12
        ],
        "energy": 6283.5,
        "AssignedTask": null
    }
3830 ],
    "totalEnergyTasks": 25134,
    "allTasks": [
        {
3835            "id": "T23",
            "required_equipment": 8,
            "energy_cost": 1500,

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    "CreatedAt": 1759044740
  },
  {
3840    "id": "T12",
    "required_equipment": 12,
    "energy_cost": 10,
    "CreatedAt": 1759044729
  },
3845  {
    "id": "T30",
    "required_equipment": 15,
    "energy_cost": 10,
    "CreatedAt": 1759044747
3850  },
  {
    "id": "T32",
    "required_equipment": 12,
    "energy_cost": 994,
3855    "CreatedAt": 1759044719
  },
  {
    "id": "T20",
    "required_equipment": 5,
3860    "energy_cost": 10,
    "CreatedAt": 1759044737
  },
  {
    "id": "T29",
3865    "required_equipment": 14,
    "energy_cost": 1500,
    "CreatedAt": 1759044746
  },
  {
3870    "id": "T25",
    "required_equipment": 10,
    "energy_cost": 1500,
    "CreatedAt": 1759044742
  },
3875  {
    "id": "T9",
    "required_equipment": 9,
    "energy_cost": 1500,
    "CreatedAt": 1759044726
3880  },
  {
    "id": "T21",
    "required_equipment": 6,
    "energy_cost": 1500,
3885    "CreatedAt": 1759044738
  },
  {
    "id": "T28",

```

```

3890     "required_equipment": 13,
        "energy_cost": 10,
        "CreatedAt": 1759044745
    },
    {
3895     "id": "T7",
        "required_equipment": 7,
        "energy_cost": 1500,
        "CreatedAt": 1759044724
    },
    {
3900     "id": "T1",
        "required_equipment": 1,
        "energy_cost": 1500,
        "CreatedAt": 1759044718
    },
    {
3905     "id": "T10",
        "required_equipment": 10,
        "energy_cost": 10,
        "CreatedAt": 1759044727
3910 },
    {
        "id": "T8",
        "required_equipment": 8,
        "energy_cost": 10,
3915     "CreatedAt": 1759044725
    },
    {
        "id": "T14",
        "required_equipment": 14,
3920     "energy_cost": 10,
        "CreatedAt": 1759044731
    },
    {
        "id": "T18",
3925     "required_equipment": 3,
        "energy_cost": 10,
        "CreatedAt": 1759044735
    },
    {
3930     "id": "T26",
        "required_equipment": 11,
        "energy_cost": 10,
        "CreatedAt": 1759044743
    },
    {
3935     "id": "T2",
        "required_equipment": 2,
        "energy_cost": 10,
        "CreatedAt": 1759044719
3940 },

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3945     {
        "id": "T4",
        "required_equipment": 4,
        "energy_cost": 10,
        "CreatedAt": 1759044721
    },
    {
3950     "id": "T3",
        "required_equipment": 3,
        "energy_cost": 1500,
        "CreatedAt": 1759044720
    },
    {
3955     "id": "T11",
        "required_equipment": 11,
        "energy_cost": 1500,
        "CreatedAt": 1759044728
    },
    {
3960     "id": "T16",
        "required_equipment": 1,
        "energy_cost": 10,
        "CreatedAt": 1759044733
    },
    {
3965     "id": "T22",
        "required_equipment": 7,
        "energy_cost": 10,
        "CreatedAt": 1759044739
3970 },
    {
        "id": "T17",
        "required_equipment": 2,
        "energy_cost": 1500,
3975     "CreatedAt": 1759044734
    },
    {
        "id": "T24",
        "required_equipment": 9,
3980     "energy_cost": 10,
        "CreatedAt": 1759044741
    },
    {
        "id": "T19",
3985     "required_equipment": 4,
        "energy_cost": 1500,
        "CreatedAt": 1759044736
    },
    {
3990     "id": "T27",
        "required_equipment": 12,
        "energy_cost": 1500,

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    "CreatedAt": 1759044744
  },
3995 {
    "id": "T13",
    "required_equipment": 13,
    "energy_cost": 1500,
    "CreatedAt": 1759044730
4000 },
    {
    "id": "T31",
    "required_equipment": 7,
    "energy_cost": 1490,
4005 "CreatedAt": 1759044718
    },
    {
    "id": "T15",
    "required_equipment": 15,
4010 "energy_cost": 1500,
    "CreatedAt": 1759044732
    },
    {
    "id": "T5",
4015 "required_equipment": 5,
    "energy_cost": 1500,
    "CreatedAt": 1759044722
    },
    {
4020 "id": "T6",
    "required_equipment": 6,
    "energy_cost": 10,
    "CreatedAt": 1759044723
    }
4025 ],
    "totalTime": "48.636µs",
    "stepMetrics": [
    {
4030 "step": 1,
    "lenTasksDone": 32,
    "tasksDone": [
    "T23",
    "T12",
    "T30",
4035 "T32",
    "T20",
    "T29",
    "T25",
    "T9",
    "T21",
4040 "T28",
    "T7",
    "T1",
    "T10",

```



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4045         "T8",
            "T14",
            "T18",
            "T26",
            "T2",
4050         "T4",
            "T3",
            "T11",
            "T16",
            "T22",
4055         "T17",
            "T24",
            "T19",
            "T27",
            "T13",
4060         "T31",
            "T15",
            "T5",
            "T6"
        ],
4065         "energyUsed": 25134,
        "lenNotExecTasks": 0,
        "realTime": "7.176µs"
    },
    {
4070         "step": 2,
        "lenTasksDone": 32,
        "tasksDone": [
            "T23",
            "T12",
4075         "T30",
            "T32",
            "T20",
            "T29",
            "T25",
4080         "T9",
            "T21",
            "T28",
            "T7",
            "T1",
4085         "T10",
            "T8",
            "T14",
            "T18",
            "T26",
4090         "T2",
            "T4",
            "T3",
            "T11",
            "T16",
4095         "T22",
            "T17",

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4100         "T24",
            "T19",
            "T27",
            "T13",
            "T31",
            "T15",
            "T5",
            "T6"
4105     ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,
        "realTime": "11.986µs"
    },
4110 {
        "step": 3,
        "lenTasksDone": 32,
        "tasksDone": [
4115         "T23",
            "T12",
            "T30",
            "T32",
            "T20",
            "T29",
4120         "T25",
            "T9",
            "T21",
            "T28",
            "T7",
4125         "T1",
            "T10",
            "T8",
            "T14",
            "T18",
4130         "T26",
            "T2",
            "T4",
            "T3",
            "T11",
4135         "T16",
            "T22",
            "T17",
            "T24",
            "T19",
4140         "T27",
            "T13",
            "T31",
            "T15",
            "T5",
4145         "T6"
        ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,

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    "realTime": "5.054µs"
4150 },
    {
        "step": 4,
        "lenTasksDone": 32,
        "tasksDone": [
4155     "T23",
        "T12",
        "T30",
        "T32",
        "T20",
4160     "T29",
        "T25",
        "T9",
        "T21",
        "T28",
4165     "T7",
        "T1",
        "T10",
        "T8",
        "T14",
4170     "T18",
        "T26",
        "T2",
        "T4",
        "T3",
4175     "T11",
        "T16",
        "T22",
        "T17",
        "T24",
4180     "T19",
        "T27",
        "T13",
        "T31",
        "T15",
4185     "T5",
        "T6"
    ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,
4190     "realTime": "11.562µs"
    },
    {
        "step": 5,
        "lenTasksDone": 32,
4195     "tasksDone": [
        "T23",
        "T12",
        "T30",
        "T32",
4200     "T20",

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```

    "T29",
    "T25",
    "T9",
    "T21",
4205    "T28",
    "T7",
    "T1",
    "T10",
    "T8",
4210    "T14",
    "T18",
    "T26",
    "T2",
    "T4",
4215    "T3",
    "T11",
    "T16",
    "T22",
    "T17",
4220    "T24",
    "T19",
    "T27",
    "T13",
    "T31",
4225    "T15",
    "T5",
    "T6"
  ],
  "energyUsed": 25134,
4230  "lenNotExecTasks": 0,
  "realTime": "9.689µs"
}
]
},
4235  "Энергосберегающая": {
    "totalTasks": 32,
    "totalMachine": 8,
    "totalEnergyMachines": 50264,
    "allMachines": [
4240      {
        "id": "M1",
        "equipment": [
          1,
          3,
4245          9
        ],
        "energy": 6283.5,
        "AssignedTask": null
      },
4250      {
        "id": "M2",
        "equipment": [

```

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    2,
    9
4255 ],
    "energy": 6283.5,
    "AssignedTask": null
  },
  {
4260   "id": "M3",
    "equipment": [
        1,
        7,
        14
4265   ],
    "energy": 6283.5,
    "AssignedTask": null
  },
  {
4270   "id": "M4",
    "equipment": [
        4,
        5,
        6,
4275   15
    ],
    "energy": 6283.5,
    "AssignedTask": null
  },
  {
4280   "id": "M5",
    "equipment": [
        11
    ],
    "energy": 6283.5,
    "AssignedTask": null
4285  },
  {
    "id": "M6",
4290   "equipment": [
        8,
        13
    ],
    "energy": 6283.5,
    "AssignedTask": null
4295  },
  {
    "id": "M7",
4300   "equipment": [
        10
    ],
    "energy": 6283.5,
    "AssignedTask": null
  },

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4305     {
        "id": "M8",
        "equipment": [
            12
        ],
4310     "energy": 6283.5,
        "AssignedTask": null
    }
],
"totalEnergyTasks": 25134,
4315 "allTasks": [
    {
        "id": "T23",
        "required_equipment": 8,
        "energy_cost": 1500,
4320     "CreatedAt": 1759044740
    },
    {
        "id": "T12",
        "required_equipment": 12,
4325     "energy_cost": 10,
        "CreatedAt": 1759044729
    },
    {
        "id": "T30",
        "required_equipment": 15,
4330     "energy_cost": 10,
        "CreatedAt": 1759044747
    },
    {
        "id": "T32",
        "required_equipment": 12,
4335     "energy_cost": 994,
        "CreatedAt": 1759044719
    },
    {
        "id": "T20",
        "required_equipment": 5,
4340     "energy_cost": 10,
        "CreatedAt": 1759044737
    },
4345     },
    {
        "id": "T29",
        "required_equipment": 14,
        "energy_cost": 1500,
4350     "CreatedAt": 1759044746
    },
    {
        "id": "T25",
        "required_equipment": 10,
4355     "energy_cost": 1500,
        "CreatedAt": 1759044742
    }
]

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},
{
  "id": "T9",
  "required_equipment": 9,
  "energy_cost": 1500,
  "CreatedAt": 1759044726
},
{
  "id": "T21",
  "required_equipment": 6,
  "energy_cost": 1500,
  "CreatedAt": 1759044738
},
{
  "id": "T28",
  "required_equipment": 13,
  "energy_cost": 10,
  "CreatedAt": 1759044745
},
{
  "id": "T7",
  "required_equipment": 7,
  "energy_cost": 1500,
  "CreatedAt": 1759044724
},
{
  "id": "T1",
  "required_equipment": 1,
  "energy_cost": 1500,
  "CreatedAt": 1759044718
},
{
  "id": "T10",
  "required_equipment": 10,
  "energy_cost": 10,
  "CreatedAt": 1759044727
},
{
  "id": "T8",
  "required_equipment": 8,
  "energy_cost": 10,
  "CreatedAt": 1759044725
},
{
  "id": "T14",
  "required_equipment": 14,
  "energy_cost": 10,
  "CreatedAt": 1759044731
},
{
  "id": "T18",
  "required_equipment": 3,

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    "energy_cost": 10,
4410    "CreatedAt": 1759044735
  },
  {
    "id": "T26",
    "required_equipment": 11,
4415    "energy_cost": 10,
    "CreatedAt": 1759044743
  },
  {
    "id": "T2",
4420    "required_equipment": 2,
    "energy_cost": 10,
    "CreatedAt": 1759044719
  },
  {
    "id": "T4",
4425    "required_equipment": 4,
    "energy_cost": 10,
    "CreatedAt": 1759044721
  },
  {
4430    "id": "T3",
    "required_equipment": 3,
    "energy_cost": 1500,
    "CreatedAt": 1759044720
4435  },
  {
    "id": "T11",
    "required_equipment": 11,
    "energy_cost": 1500,
4440    "CreatedAt": 1759044728
  },
  {
    "id": "T16",
4445    "required_equipment": 1,
    "energy_cost": 10,
    "CreatedAt": 1759044733
  },
  {
    "id": "T22",
4450    "required_equipment": 7,
    "energy_cost": 10,
    "CreatedAt": 1759044739
  },
  {
4455    "id": "T17",
    "required_equipment": 2,
    "energy_cost": 1500,
    "CreatedAt": 1759044734
  },
4460  {

```



```

    "id": "T24",
    "required_equipment": 9,
    "energy_cost": 10,
    "CreatedAt": 1759044741
4465 },
    {
    "id": "T19",
    "required_equipment": 4,
    "energy_cost": 1500,
4470 "CreatedAt": 1759044736
    },
    {
    "id": "T27",
    "required_equipment": 12,
    "energy_cost": 1500,
4475 "CreatedAt": 1759044744
    },
    {
    "id": "T13",
    "required_equipment": 13,
    "energy_cost": 1500,
4480 "CreatedAt": 1759044730
    },
    {
4485 "id": "T31",
    "required_equipment": 7,
    "energy_cost": 1490,
    "CreatedAt": 1759044718
    },
4490 {
    "id": "T15",
    "required_equipment": 15,
    "energy_cost": 1500,
    "CreatedAt": 1759044732
4495 },
    {
    "id": "T5",
    "required_equipment": 5,
    "energy_cost": 1500,
4500 "CreatedAt": 1759044722
    },
    {
    "id": "T6",
    "required_equipment": 6,
    "energy_cost": 10,
4505 "CreatedAt": 1759044723
    }
  ],
  "totalTime": "50.776µs",
4510 "stepMetrics": [
    {
    "step": 1,

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4515     "lenTasksDone": 32,
        "tasksDone": [
            "T18",
            "T2",
            "T14",
            "T30",
            "T26",
4520     "T28",
            "T10",
            "T12",
            "T16",
            "T24",
4525     "T22",
            "T20",
            "T11",
            "T8",
            "T25",
4530     "T32",
            "T9",
            "T17",
            "T31",
            "T4",
4535     "T23",
            "T27",
            "T1",
            "T29",
            "T6",
4540     "T13",
            "T3",
            "T7",
            "T21",
            "T19",
4545     "T15",
            "T5"
        ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,
4550     "realTime": "10.425µs"
    },
    {
        "step": 2,
        "lenTasksDone": 32,
4555     "tasksDone": [
            "T18",
            "T2",
            "T14",
            "T30",
            "T26",
4560     "T28",
            "T10",
            "T12",
            "T16",

```

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4565         "T24",
            "T22",
            "T20",
            "T11",
            "T8",
4570         "T25",
            "T32",
            "T9",
            "T17",
            "T31",
4575         "T4",
            "T23",
            "T27",
            "T1",
            "T29",
4580         "T6",
            "T13",
            "T3",
            "T7",
            "T21",
4585         "T19",
            "T15",
            "T5"
        ],
        "energyUsed": 25134,
4590         "lenNotExecTasks": 0,
        "realTime": "7.689µs"
    },
    {
4595         "step": 3,
        "lenTasksDone": 32,
        "tasksDone": [
            "T18",
            "T2",
            "T14",
4600         "T30",
            "T26",
            "T28",
            "T10",
            "T12",
            "T16",
4605         "T24",
            "T22",
            "T20",
            "T11",
            "T8",
4610         "T25",
            "T32",
            "T9",
            "T17",
            "T31",
4615         "T4",

```

```

4620         "T23",
            "T27",
            "T1",
            "T29",
            "T6",
            "T13",
            "T3",
            "T7",
4625         "T21",
            "T19",
            "T15",
            "T5"
        ],
4630         "energyUsed": 25134,
        "lenNotExecTasks": 0,
        "realTime": "7.575µs"
    },
    {
4635         "step": 4,
        "lenTasksDone": 32,
        "tasksDone": [
            "T18",
            "T2",
4640         "T14",
            "T30",
            "T26",
            "T28",
            "T10",
4645         "T12",
            "T16",
            "T24",
            "T22",
            "T20",
4650         "T11",
            "T8",
            "T25",
            "T32",
            "T9",
4655         "T17",
            "T31",
            "T4",
            "T23",
            "T27",
4660         "T1",
            "T29",
            "T6",
            "T13",
            "T3",
4665         "T7",
            "T21",
            "T19",
            "T15",

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```

4670         "T5"
        ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,
        "realTime": "10.897µs"
    },
4675     {
        "step": 5,
        "lenTasksDone": 32,
        "tasksDone": [
4680             "T18",
            "T2",
            "T14",
            "T30",
            "T26",
4685             "T28",
            "T10",
            "T12",
            "T16",
            "T24",
            "T22",
4690             "T20",
            "T11",
            "T8",
            "T25",
            "T32",
4695             "T9",
            "T17",
            "T31",
            "T4",
            "T23",
            "T27",
4700             "T1",
            "T29",
            "T6",
            "T13",
            "T3",
4705             "T7",
            "T21",
            "T19",
            "T15",
            "T5"
4710        ],
        "energyUsed": 25134,
        "lenNotExecTasks": 0,
        "realTime": "11.347µs"
4715    }
    ]
}
},
4720 "min": {
    "RL (Q-learning)": {

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"totalTasks": 32,
"totalMachine": 8,
"totalEnergyMachines": 12560,
"allMachines": [
4725   {
        "id": "M1",
        "equipment": [
            1,
            3,
4730         9
        ],
        "energy": 1570.875,
        "AssignedTask": null
    },
4735   {
        "id": "M2",
        "equipment": [
            2,
            9
4740         ],
        "energy": 1570.875,
        "AssignedTask": null
    },
    {
4745     "id": "M3",
        "equipment": [
            1,
            7,
            14
4750         ],
        "energy": 1570.875,
        "AssignedTask": null
    },
    {
4755     "id": "M4",
        "equipment": [
            4,
            5,
            6,
4760         15
        ],
        "energy": 1570.875,
        "AssignedTask": null
    },
4765   {
        "id": "M5",
        "equipment": [
            11
            ],
            ],
4770     "energy": 1570.875,
        "AssignedTask": null
    },
    },

```

```

4775     {
        "id": "M6",
        "equipment": [
            8,
            13
        ],
        "energy": 1570.875,
4780     "AssignedTask": null
    },
    {
        "id": "M7",
        "equipment": [
4785     10
        ],
        "energy": 1570.875,
        "AssignedTask": null
    },
4790     {
        "id": "M8",
        "equipment": [
            12
        ],
4795     "energy": 1570.875,
        "AssignedTask": null
    }
],
"totalEnergyTasks": 25134,
4800 "allTasks": [
    {
        "id": "T23",
        "required_equipment": 8,
        "energy_cost": 1500,
4805     "CreatedAt": 1759044740
    },
    {
        "id": "T12",
        "required_equipment": 12,
4810     "energy_cost": 10,
        "CreatedAt": 1759044729
    },
    {
        "id": "T30",
        "required_equipment": 15,
4815     "energy_cost": 10,
        "CreatedAt": 1759044747
    },
    {
4820     "id": "T32",
        "required_equipment": 12,
        "energy_cost": 994,
        "CreatedAt": 1759044719
    },
    {

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```

4825     {
        "id": "T20",
        "required_equipment": 5,
        "energy_cost": 10,
        "CreatedAt": 1759044737
4830     },
        {
        "id": "T29",
        "required_equipment": 14,
        "energy_cost": 1500,
4835     "CreatedAt": 1759044746
        },
        {
        "id": "T25",
        "required_equipment": 10,
4840     "energy_cost": 1500,
        "CreatedAt": 1759044742
        },
        {
        "id": "T9",
4845     "required_equipment": 9,
        "energy_cost": 1500,
        "CreatedAt": 1759044726
        },
        {
4850     "id": "T21",
        "required_equipment": 6,
        "energy_cost": 1500,
        "CreatedAt": 1759044738
        },
4855     {
        "id": "T28",
        "required_equipment": 13,
        "energy_cost": 10,
        "CreatedAt": 1759044745
4860     },
        {
        "id": "T7",
        "required_equipment": 7,
        "energy_cost": 1500,
4865     "CreatedAt": 1759044724
        },
        {
        "id": "T1",
        "required_equipment": 1,
4870     "energy_cost": 1500,
        "CreatedAt": 1759044718
        },
        {
        "id": "T10",
4875     "required_equipment": 10,
        "energy_cost": 10,

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```

    "CreatedAt": 1759044727
  },
  {
4880    "id": "T8",
    "required_equipment": 8,
    "energy_cost": 10,
    "CreatedAt": 1759044725
  },
4885  {
    "id": "T14",
    "required_equipment": 14,
    "energy_cost": 10,
    "CreatedAt": 1759044731
4890  },
  {
    "id": "T18",
    "required_equipment": 3,
    "energy_cost": 10,
4895    "CreatedAt": 1759044735
  },
  {
    "id": "T26",
    "required_equipment": 11,
4900    "energy_cost": 10,
    "CreatedAt": 1759044743
  },
  {
    "id": "T2",
4905    "required_equipment": 2,
    "energy_cost": 10,
    "CreatedAt": 1759044719
  },
  {
4910    "id": "T4",
    "required_equipment": 4,
    "energy_cost": 10,
    "CreatedAt": 1759044721
  },
4915  {
    "id": "T3",
    "required_equipment": 3,
    "energy_cost": 1500,
    "CreatedAt": 1759044720
4920  },
  {
    "id": "T11",
    "required_equipment": 11,
    "energy_cost": 1500,
4925    "CreatedAt": 1759044728
  },
  {
    "id": "T16",

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```

4930         "required_equipment": 1,
         "energy_cost": 10,
         "CreatedAt": 1759044733
    },
    {
4935         "id": "T22",
         "required_equipment": 7,
         "energy_cost": 10,
         "CreatedAt": 1759044739
    },
    {
4940         "id": "T17",
         "required_equipment": 2,
         "energy_cost": 1500,
         "CreatedAt": 1759044734
    },
    {
4945         "id": "T24",
         "required_equipment": 9,
         "energy_cost": 10,
         "CreatedAt": 1759044741
4950     },
    {
         "id": "T19",
         "required_equipment": 4,
         "energy_cost": 1500,
4955         "CreatedAt": 1759044736
    },
    {
         "id": "T27",
4960         "required_equipment": 12,
         "energy_cost": 1500,
         "CreatedAt": 1759044744
    },
    {
         "id": "T13",
4965         "required_equipment": 13,
         "energy_cost": 1500,
         "CreatedAt": 1759044730
    },
    {
4970         "id": "T31",
         "required_equipment": 7,
         "energy_cost": 1490,
         "CreatedAt": 1759044718
    },
    {
4975         "id": "T15",
         "required_equipment": 15,
         "energy_cost": 1500,
         "CreatedAt": 1759044732
4980     },

```

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4985     {
        "id": "T5",
        "required_equipment": 5,
        "energy_cost": 1500,
        "CreatedAt": 1759044722
    },
    {
4990     "id": "T6",
        "required_equipment": 6,
        "energy_cost": 10,
        "CreatedAt": 1759044723
    }
],
4995 "totalTime": "136.962μs",
    "stepMetrics": [
        {
            "step": 1,
            "lenTasksDone": 20,
            "tasksDone": [
5000                "T18",
                    "T24",
                    "T16",
                    "T30",
                    "T26",
5005                "T8",
                    "T10",
                    "T12",
                    "T3",
                    "T9",
5010                "T1",
                    "T20",
                    "T11",
                    "T23",
                    "T25",
5015                "T32",
                    "T2",
                    "T14",
                    "T5",
                    "T28"
            ],
5020            "energyUsed": 12384.375,
            "lenNotExecTasks": 12,
            "realTime": "30.065μs"
        },
5025     {
            "step": 2,
            "lenTasksDone": 21,
            "tasksDone": [
5030                "T18",
                    "T24",
                    "T16",
                    "T20",

```

```

5035         "T26",
            "T8",
            "T10",
            "T12",
            "T1",
            "T9",
            "T14",
5040         "T5",
            "T11",
            "T23",
            "T25",
            "T32",
5045         "T2",
            "T29",
            "T30",
            "T28",
            "T22"
5050     ],
        "energyUsed": 12445.25,
        "lenNotExecTasks": 11,
        "realTime": "27.467μs"
    },
5055     {
        "step": 3,
        "lenTasksDone": 21,
        "tasksDone": [
5060             "T16",
            "T24",
            "T14",
            "T20",
            "T26",
            "T8",
5065             "T10",
            "T12",
            "T1",
            "T9",
            "T29",
5070             "T5",
            "T11",
            "T23",
            "T25",
            "T32",
5075             "T18",
            "T2",
            "T22",
            "T30",
            "T28"
5080     ],
        "energyUsed": 12445.25,
        "lenNotExecTasks": 11,
        "realTime": "24.299μs"
    },

```

```

5085 {
    "step": 4,
    "lenTasksDone": 21,
    "tasksDone": [
5090     "T16",
        "T24",
        "T14",
        "T20",
        "T26",
5095     "T8",
        "T10",
        "T12",
        "T1",
        "T9",
        "T29",
5100     "T5",
        "T11",
        "T23",
        "T25",
        "T32",
5105     "T18",
        "T2",
        "T22",
        "T30",
        "T28"
5110     ],
    "energyUsed": 12445.25,
    "lenNotExecTasks": 11,
    "realTime": "28.972µs"
5115 },
{
    "step": 5,
    "lenTasksDone": 21,
    "tasksDone": [
5120     "T16",
        "T24",
        "T14",
        "T30",
        "T26",
5125     "T8",
        "T10",
        "T12",
        "T1",
        "T9",
        "T29",
5130     "T20",
        "T11",
        "T23",
        "T25",
        "T32",
5135     "T18",
        "T2",

```

```

        "T22",
        "T5",
        "T28"
5140     ],
        "energyUsed": 12445.25,
        "lenNotExecTasks": 11,
        "realTime": "22.788µs"
    }
5145 ]
},
Минимизация" Вр3": {
    "totalTasks": 32,
    "totalMachine": 8,
5150    "totalEnergyMachines": 12560,
    "allMachines": [
        {
            "id": "M1",
            "equipment": [
5155         1,
            3,
            9
        ],
        "energy": 1570.875,
5160    "AssignedTask": null
    },
    {
        "id": "M2",
        "equipment": [
5165         2,
            9
        ],
        "energy": 1570.875,
        "AssignedTask": null
5170    },
    {
        "id": "M3",
        "equipment": [
5175         1,
            7,
            14
        ],
        "energy": 1570.875,
        "AssignedTask": null
5180    },
    {
        "id": "M4",
        "equipment": [
5185         4,
            5,
            6,
            15
        ],
    ],

```

```

5190         "energy": 1570.875,
        "AssignedTask": null
    },
    {
        "id": "M5",
5195     "equipment": [
        11
    ],
        "energy": 1570.875,
        "AssignedTask": null
    },
5200     {
        "id": "M6",
        "equipment": [
            8,
            13
5205     ],
        "energy": 1570.875,
        "AssignedTask": null
    },
    {
5210     "id": "M7",
        "equipment": [
            10
        ],
        "energy": 1570.875,
5215     "AssignedTask": null
    },
    {
        "id": "M8",
5220     "equipment": [
            12
        ],
        "energy": 1570.875,
        "AssignedTask": null
    }
5225 ],
    "totalEnergyTasks": 25134,
    "allTasks": [
        {
5230     "id": "T23",
        "required_equipment": 8,
        "energy_cost": 1500,
        "CreatedAt": 1759044740
        },
        {
5235     "id": "T12",
        "required_equipment": 12,
        "energy_cost": 10,
        "CreatedAt": 1759044729
        },
5240     {

```

```

5245     "id": "T30",
        "required_equipment": 15,
        "energy_cost": 10,
        "CreatedAt": 1759044747
    },
    {
        "id": "T32",
        "required_equipment": 12,
5250     "energy_cost": 994,
        "CreatedAt": 1759044719
    },
    {
        "id": "T20",
        "required_equipment": 5,
5255     "energy_cost": 10,
        "CreatedAt": 1759044737
    },
    {
        "id": "T29",
5260     "required_equipment": 14,
        "energy_cost": 1500,
        "CreatedAt": 1759044746
    },
    {
5265     "id": "T25",
        "required_equipment": 10,
        "energy_cost": 1500,
        "CreatedAt": 1759044742
    },
5270     {
        "id": "T9",
        "required_equipment": 9,
        "energy_cost": 1500,
        "CreatedAt": 1759044726
5275     },
    {
        "id": "T21",
        "required_equipment": 6,
5280     "energy_cost": 1500,
        "CreatedAt": 1759044738
    },
    {
        "id": "T28",
5285     "required_equipment": 13,
        "energy_cost": 10,
        "CreatedAt": 1759044745
    },
    {
5290     "id": "T7",
        "required_equipment": 7,
        "energy_cost": 1500,
        "CreatedAt": 1759044724

```



```

},
{
5295   "id": "T1",
      "required_equipment": 1,
      "energy_cost": 1500,
      "CreatedAt": 1759044718
},
5300 {
      "id": "T10",
      "required_equipment": 10,
      "energy_cost": 10,
      "CreatedAt": 1759044727
5305 },
{
      "id": "T8",
      "required_equipment": 8,
      "energy_cost": 10,
5310 "CreatedAt": 1759044725
},
{
      "id": "T14",
      "required_equipment": 14,
5315 "energy_cost": 10,
      "CreatedAt": 1759044731
},
{
      "id": "T18",
5320 "required_equipment": 3,
      "energy_cost": 10,
      "CreatedAt": 1759044735
},
{
5325 "id": "T26",
      "required_equipment": 11,
      "energy_cost": 10,
      "CreatedAt": 1759044743
},
5330 {
      "id": "T2",
      "required_equipment": 2,
      "energy_cost": 10,
      "CreatedAt": 1759044719
5335 },
{
      "id": "T4",
      "required_equipment": 4,
      "energy_cost": 10,
5340 "CreatedAt": 1759044721
},
{
      "id": "T3",
      "required_equipment": 3,

```

```

5345         "energy_cost": 1500,
           "CreatedAt": 1759044720
        },
        {
5350         "id": "T11",
           "required_equipment": 11,
           "energy_cost": 1500,
           "CreatedAt": 1759044728
        },
        {
5355         "id": "T16",
           "required_equipment": 1,
           "energy_cost": 10,
           "CreatedAt": 1759044733
        },
        {
5360         "id": "T22",
           "required_equipment": 7,
           "energy_cost": 10,
           "CreatedAt": 1759044739
5365     },
        {
           "id": "T17",
           "required_equipment": 2,
           "energy_cost": 1500,
5370         "CreatedAt": 1759044734
        },
        {
           "id": "T24",
           "required_equipment": 9,
5375         "energy_cost": 10,
           "CreatedAt": 1759044741
        },
        {
           "id": "T19",
5380         "required_equipment": 4,
           "energy_cost": 1500,
           "CreatedAt": 1759044736
        },
        {
5385         "id": "T27",
           "required_equipment": 12,
           "energy_cost": 1500,
           "CreatedAt": 1759044744
        },
        {
5390         "id": "T13",
           "required_equipment": 13,
           "energy_cost": 1500,
           "CreatedAt": 1759044730
5395     },
        {

```

```

5400     "id": "T31",
        "required_equipment": 7,
        "energy_cost": 1490,
        "CreatedAt": 1759044718
    },
    {
5405     "id": "T15",
        "required_equipment": 15,
        "energy_cost": 1500,
        "CreatedAt": 1759044732
    },
    {
5410     "id": "T5",
        "required_equipment": 5,
        "energy_cost": 1500,
        "CreatedAt": 1759044722
    },
    {
5415     "id": "T6",
        "required_equipment": 6,
        "energy_cost": 10,
        "CreatedAt": 1759044723
    }
5420 ],
    "totalTime": "65.887µs",
    "stepMetrics": [
        {
5425     "step": 1,
        "lenTasksDone": 18,
        "tasksDone": [
            "T23",
            "T12",
            "T30",
5430     "T29",
            "T25",
            "T9",
            "T26",
            "T32",
5435     "T20",
            "T28",
            "T10",
            "T2",
            "T11",
5440     "T21",
            "T8",
            "T17",
            "T4",
            "T24"
5445 ],
        "energyUsed": 12394.375,
        "lenNotExecTasks": 14,
        "realTime": "13.856µs"

```

```

5450     },
    {
        "step": 2,
        "lenTasksDone": 18,
        "tasksDone": [
5455             "T23",
             "T12",
             "T30",
             "T29",
             "T25",
5460             "T9",
             "T26",
             "T32",
             "T20",
             "T28",
5465             "T10",
             "T2",
             "T11",
             "T21",
             "T8",
5470             "T17",
             "T4",
             "T24"
        ],
        "energyUsed": 12394.375,
        "lenNotExecTasks": 14,
5475        "realTime": "11.274µs"
    },
    {
        "step": 3,
        "lenTasksDone": 18,
5480        "tasksDone": [
             "T23",
             "T12",
             "T30",
             "T29",
5485             "T25",
             "T9",
             "T26",
             "T32",
             "T20",
5490             "T28",
             "T10",
             "T2",
             "T11",
             "T21",
5495             "T8",
             "T17",
             "T4",
             "T24"
        ],
5500        "energyUsed": 12394.375,

```

```

    "lenNotExecTasks": 14,
    "realTime": "16.127µs"
  },
  {
5505    "step": 4,
    "lenTasksDone": 18,
    "tasksDone": [
5510      "T23",
      "T12",
      "T30",
      "T29",
      "T25",
      "T9",
      "T26",
5515      "T32",
      "T20",
      "T28",
      "T10",
      "T2",
5520      "T11",
      "T21",
      "T8",
      "T17",
      "T4",
5525      "T24"
    ],
    "energyUsed": 12394.375,
    "lenNotExecTasks": 14,
    "realTime": "9.481µs"
5530  },
  {
    "step": 5,
    "lenTasksDone": 18,
    "tasksDone": [
5535      "T23",
      "T12",
      "T30",
      "T29",
      "T25",
      "T9",
      "T26",
5540      "T32",
      "T20",
      "T28",
      "T10",
      "T2",
5545      "T11",
      "T21",
      "T8",
      "T17",
      "T4",
5550      "T24"
    ]
  }

```

```

    ],
    "energyUsed": 12394.375,
5555    "lenNotExecTasks": 14,
    "realTime": "9.269µs"
  }
]
},
5560 Прямого" перебора": {
  "totalTasks": 32,
  "totalMachine": 8,
  "totalEnergyMachines": 12560,
  "allMachines": [
5565    {
      "id": "M1",
      "equipment": [
        1,
        3,
5570        9
      ],
      "energy": 1570.875,
      "AssignedTask": null
    },
5575    {
      "id": "M2",
      "equipment": [
        2,
        9
5580      ],
      "energy": 1570.875,
      "AssignedTask": null
    },
    {
5585      "id": "M3",
      "equipment": [
        1,
        7,
        14
5590      ],
      "energy": 1570.875,
      "AssignedTask": null
    },
    {
5595      "id": "M4",
      "equipment": [
        4,
        5,
        6,
5600        15
      ],
      "energy": 1570.875,
      "AssignedTask": null
    },
  ],

```

```

5605     {
        "id": "M5",
        "equipment": [
5610             11
        ],
        "energy": 1570.875,
        "AssignedTask": null
    },
    {
        "id": "M6",
5615         "equipment": [
            8,
            13
        ],
        "energy": 1570.875,
5620         "AssignedTask": null
    },
    {
        "id": "M7",
5625         "equipment": [
            10
        ],
        "energy": 1570.875,
        "AssignedTask": null
    },
5630     {
        "id": "M8",
        "equipment": [
            12
        ],
5635         "energy": 1570.875,
        "AssignedTask": null
    }
],
"totalEnergyTasks": 25134,
5640 "allTasks": [
    {
        "id": "T23",
        "required_equipment": 8,
        "energy_cost": 1500,
5645         "CreatedAt": 1759044740
    },
    {
        "id": "T12",
        "required_equipment": 12,
5650         "energy_cost": 10,
        "CreatedAt": 1759044729
    },
    {
        "id": "T30",
5655         "required_equipment": 15,
        "energy_cost": 10,

```

```

    "CreatedAt": 1759044747
  },
  {
5660    "id": "T32",
      "required_equipment": 12,
      "energy_cost": 994,
      "CreatedAt": 1759044719
    },
5665    {
      "id": "T20",
      "required_equipment": 5,
      "energy_cost": 10,
      "CreatedAt": 1759044737
5670    },
    {
      "id": "T29",
      "required_equipment": 14,
      "energy_cost": 1500,
5675      "CreatedAt": 1759044746
    },
    {
      "id": "T25",
      "required_equipment": 10,
5680      "energy_cost": 1500,
      "CreatedAt": 1759044742
    },
    {
      "id": "T9",
5685      "required_equipment": 9,
      "energy_cost": 1500,
      "CreatedAt": 1759044726
    },
    {
5690      "id": "T21",
      "required_equipment": 6,
      "energy_cost": 1500,
      "CreatedAt": 1759044738
    },
5695    {
      "id": "T28",
      "required_equipment": 13,
      "energy_cost": 10,
      "CreatedAt": 1759044745
5700    },
    {
      "id": "T7",
      "required_equipment": 7,
      "energy_cost": 1500,
5705      "CreatedAt": 1759044724
    },
    {
      "id": "T1",

```



```

5710         "required_equipment": 1,
           "energy_cost": 1500,
           "CreatedAt": 1759044718
5715     },
           {
           "id": "T10",
           "required_equipment": 10,
           "energy_cost": 10,
           "CreatedAt": 1759044727
5720     },
           {
           "id": "T8",
           "required_equipment": 8,
           "energy_cost": 10,
           "CreatedAt": 1759044725
5725     },
           {
           "id": "T14",
           "required_equipment": 14,
           "energy_cost": 10,
           "CreatedAt": 1759044731
5730     },
           {
           "id": "T18",
           "required_equipment": 3,
           "energy_cost": 10,
           "CreatedAt": 1759044735
5735     },
           {
           "id": "T26",
           "required_equipment": 11,
           "energy_cost": 10,
           "CreatedAt": 1759044743
5740     },
           {
           "id": "T2",
           "required_equipment": 2,
           "energy_cost": 10,
           "CreatedAt": 1759044719
5745     },
           {
           "id": "T4",
           "required_equipment": 4,
           "energy_cost": 10,
           "CreatedAt": 1759044721
5750     },
           {
           "id": "T3",
           "required_equipment": 3,
           "energy_cost": 1500,
           "CreatedAt": 1759044720
5755     },
           {
           "id": "T3",
           "required_equipment": 3,
           "energy_cost": 1500,
           "CreatedAt": 1759044720
5760     },
           {
           "id": "T3",
           "required_equipment": 3,
           "energy_cost": 1500,
           "CreatedAt": 1759044720

```

```

5765 {
      "id": "T11",
      "required_equipment": 11,
      "energy_cost": 1500,
      "CreatedAt": 1759044728
    },
5770 {
      "id": "T16",
      "required_equipment": 1,
      "energy_cost": 10,
      "CreatedAt": 1759044733
    },
5775 {
      "id": "T22",
      "required_equipment": 7,
      "energy_cost": 10,
      "CreatedAt": 1759044739
    },
5780 {
      "id": "T17",
      "required_equipment": 2,
      "energy_cost": 1500,
      "CreatedAt": 1759044734
    },
5785 {
      "id": "T24",
      "required_equipment": 9,
      "energy_cost": 10,
      "CreatedAt": 1759044741
    },
5790 {
      "id": "T19",
      "required_equipment": 4,
      "energy_cost": 1500,
      "CreatedAt": 1759044736
    },
5795 {
      "id": "T27",
      "required_equipment": 12,
      "energy_cost": 1500,
      "CreatedAt": 1759044744
    },
5800 {
      "id": "T13",
      "required_equipment": 13,
      "energy_cost": 1500,
      "CreatedAt": 1759044730
    },
5805 {
      "id": "T31",
      "required_equipment": 7,
      "energy_cost": 1490,

```

```

    "CreatedAt": 1759044718
  },
5815 {
    "id": "T15",
    "required_equipment": 15,
    "energy_cost": 1500,
    "CreatedAt": 1759044732
5820 },
  {
    "id": "T5",
    "required_equipment": 5,
    "energy_cost": 1500,
5825 "CreatedAt": 1759044722
  },
  {
    "id": "T6",
    "required_equipment": 6,
5830 "energy_cost": 10,
    "CreatedAt": 1759044723
  }
],
"totalTime": "22.434µs",
5835 "stepMetrics": [
  {
    "step": 1,
    "lenTasksDone": 18,
    "tasksDone": [
5840 "T9",
    "T2",
    "T29",
    "T30",
    "T26",
5845 "T23",
    "T25",
    "T12",
    "T17",
    "T20",
5850 "T11",
    "T28",
    "T10",
    "T32",
    "T24",
5855 "T21",
    "T8",
    "T4"
  ],
  "energyUsed": 12394.375,
5860 "lenNotExecTasks": 14,
  "realTime": "5.785µs"
},
{
  "step": 2,

```

```

5865     "lenTasksDone": 18,
        "tasksDone": [
            "T9",
            "T2",
            "T29",
5870     "T30",
            "T26",
            "T23",
            "T25",
            "T12",
5875     "T17",
            "T20",
            "T11",
            "T28",
            "T10",
5880     "T32",
            "T24",
            "T21",
            "T8",
            "T4"
5885     ],
        "energyUsed": 12394.375,
        "lenNotExecTasks": 14,
        "realTime": "2.201µs"
5890     },
    {
        "step": 3,
        "lenTasksDone": 18,
        "tasksDone": [
            "T9",
5895     "T2",
            "T29",
            "T30",
            "T26",
            "T23",
5900     "T25",
            "T12",
            "T17",
            "T20",
            "T11",
5905     "T28",
            "T10",
            "T32",
            "T24",
            "T21",
5910     "T8",
            "T4"
        ],
        "energyUsed": 12394.375,
        "lenNotExecTasks": 14,
5915     "realTime": "1.403µs"
    },

```

```

5920 {
    "step": 4,
    "lenTasksDone": 18,
    "tasksDone": [
        "T9",
        "T2",
        "T29",
        "T30",
5925 "T26",
        "T23",
        "T25",
        "T12",
        "T17",
5930 "T20",
        "T11",
        "T28",
        "T10",
        "T32",
5935 "T24",
        "T21",
        "T8",
        "T4"
    ],
5940 "energyUsed": 12394.375,
    "lenNotExecTasks": 14,
    "realTime": "1.444µs"
},
5945 {
    "step": 5,
    "lenTasksDone": 18,
    "tasksDone": [
        "T9",
        "T2",
5950 "T29",
        "T30",
        "T26",
        "T23",
        "T25",
5955 "T12",
        "T17",
        "T20",
        "T11",
        "T28",
5960 "T10",
        "T32",
        "T24",
        "T21",
        "T8",
5965 "T4"
    ],
    "energyUsed": 12394.375,
    "lenNotExecTasks": 14,

```

```

    "realTime": "1.147µs"
5970   }
      ]
    },
    Рыночных" методов": {
      "totalTasks": 32,
5975   "totalMachine": 8,
      "totalEnergyMachines": 12560,
      "allMachines": [
        {
5980   "id": "M1",
          "equipment": [
            1,
            3,
            9
          ],
5985   "energy": 1570.875,
          "AssignedTask": null
        },
        {
5990   "id": "M2",
          "equipment": [
            2,
            9
          ],
          "energy": 1570.875,
5995   "AssignedTask": null
        },
        {
          "id": "M3",
          "equipment": [
6000   1,
            7,
            14
          ],
          "energy": 1570.875,
6005   "AssignedTask": null
        },
        {
          "id": "M4",
          "equipment": [
6010   4,
            5,
            6,
            15
          ],
          "energy": 1570.875,
6015   "AssignedTask": null
        },
        {
          "id": "M5",
6020   "equipment": [

```

```

        11
    ],
    "energy": 1570.875,
    "AssignedTask": null
6025 },
    {
        "id": "M6",
        "equipment": [
6030     8,
        13
        ],
        "energy": 1570.875,
        "AssignedTask": null
6035 },
    {
        "id": "M7",
        "equipment": [
        10
6040     ],
        "energy": 1570.875,
        "AssignedTask": null
    },
    {
6045     "id": "M8",
        "equipment": [
        12
        ],
        "energy": 1570.875,
        "AssignedTask": null
6050 }
    ],
    "totalEnergyTasks": 25134,
    "allTasks": [
6055     {
        "id": "T23",
        "required_equipment": 8,
        "energy_cost": 1500,
        "CreatedAt": 1759044740
    },
6060     {
        "id": "T12",
        "required_equipment": 12,
        "energy_cost": 10,
        "CreatedAt": 1759044729
6065     },
    {
        "id": "T30",
        "required_equipment": 15,
        "energy_cost": 10,
6070     "CreatedAt": 1759044747
    },
    {

```

```

6075     "id": "T32",
        "required_equipment": 12,
        "energy_cost": 994,
        "CreatedAt": 1759044719
    },
    {
6080     "id": "T20",
        "required_equipment": 5,
        "energy_cost": 10,
        "CreatedAt": 1759044737
    },
    {
6085     "id": "T29",
        "required_equipment": 14,
        "energy_cost": 1500,
        "CreatedAt": 1759044746
    },
    {
6090     "id": "T25",
        "required_equipment": 10,
        "energy_cost": 1500,
        "CreatedAt": 1759044742
6095 },
    {
        "id": "T9",
        "required_equipment": 9,
        "energy_cost": 1500,
6100     "CreatedAt": 1759044726
    },
    {
        "id": "T21",
        "required_equipment": 6,
6105     "energy_cost": 1500,
        "CreatedAt": 1759044738
    },
    {
6110     "id": "T28",
        "required_equipment": 13,
        "energy_cost": 10,
        "CreatedAt": 1759044745
    },
    {
6115     "id": "T7",
        "required_equipment": 7,
        "energy_cost": 1500,
        "CreatedAt": 1759044724
    },
6120     {
        "id": "T1",
        "required_equipment": 1,
        "energy_cost": 1500,
        "CreatedAt": 1759044718
    }

```



```

6125     },
        {
            "id": "T10",
            "required_equipment": 10,
            "energy_cost": 10,
6130     "CreatedAt": 1759044727
        },
        {
            "id": "T8",
            "required_equipment": 8,
6135     "energy_cost": 10,
            "CreatedAt": 1759044725
        },
        {
            "id": "T14",
6140     "required_equipment": 14,
            "energy_cost": 10,
            "CreatedAt": 1759044731
        },
        {
6145     "id": "T18",
            "required_equipment": 3,
            "energy_cost": 10,
            "CreatedAt": 1759044735
        },
6150     {
            "id": "T26",
            "required_equipment": 11,
            "energy_cost": 10,
            "CreatedAt": 1759044743
6155     },
        {
            "id": "T2",
            "required_equipment": 2,
            "energy_cost": 10,
6160     "CreatedAt": 1759044719
        },
        {
            "id": "T4",
            "required_equipment": 4,
6165     "energy_cost": 10,
            "CreatedAt": 1759044721
        },
        {
            "id": "T3",
6170     "required_equipment": 3,
            "energy_cost": 1500,
            "CreatedAt": 1759044720
        },
6175     {
            "id": "T11",
            "required_equipment": 11,

```

```

        "energy_cost": 1500,
        "CreatedAt": 1759044728
    },
6180    {
        "id": "T16",
        "required_equipment": 1,
        "energy_cost": 10,
        "CreatedAt": 1759044733
6185    },
    {
        "id": "T22",
        "required_equipment": 7,
        "energy_cost": 10,
6190    "CreatedAt": 1759044739
    },
    {
        "id": "T17",
        "required_equipment": 2,
6195    "energy_cost": 1500,
        "CreatedAt": 1759044734
    },
    {
        "id": "T24",
6200    "required_equipment": 9,
        "energy_cost": 10,
        "CreatedAt": 1759044741
    },
    {
6205    "id": "T19",
        "required_equipment": 4,
        "energy_cost": 1500,
        "CreatedAt": 1759044736
    },
6210    {
        "id": "T27",
        "required_equipment": 12,
        "energy_cost": 1500,
        "CreatedAt": 1759044744
6215    },
    {
        "id": "T13",
        "required_equipment": 13,
        "energy_cost": 1500,
6220    "CreatedAt": 1759044730
    },
    {
        "id": "T31",
        "required_equipment": 7,
6225    "energy_cost": 1490,
        "CreatedAt": 1759044718
    },
    {

```

```

6230         "id": "T15",
        "required_equipment": 15,
        "energy_cost": 1500,
        "CreatedAt": 1759044732
    },
    {
6235         "id": "T5",
        "required_equipment": 5,
        "energy_cost": 1500,
        "CreatedAt": 1759044722
    },
6240     {
        "id": "T6",
        "required_equipment": 6,
        "energy_cost": 10,
        "CreatedAt": 1759044723
6245     }
    ],
    "totalTime": "34.114µs",
    "stepMetrics": [
        {
6250         "step": 1,
        "lenTasksDone": 18,
        "tasksDone": [
            "T23",
            "T12",
6255         "T30",
            "T32",
            "T20",
            "T29",
            "T25",
6260         "T9",
            "T21",
            "T28",
            "T10",
            "T8",
6265         "T26",
            "T2",
            "T4",
            "T11",
            "T17",
6270         "T24"
        ],
        "energyUsed": 12394.375,
        "lenNotExecTasks": 14,
        "realTime": "5.314µs"
6275     },
    {
        "step": 2,
        "lenTasksDone": 18,
        "tasksDone": [
6280         "T23",

```

```

6285     "T12",
        "T30",
        "T32",
        "T20",
        "T29",
        "T25",
        "T9",
        "T21",
        "T28",
6290     "T10",
        "T8",
        "T26",
        "T2",
        "T4",
6295     "T11",
        "T17",
        "T24"
    ],
    "energyUsed": 12394.375,
6300     "lenNotExecTasks": 14,
    "realTime": "4.793μs"
},
{
6305     "step": 3,
    "lenTasksDone": 18,
    "tasksDone": [
        "T23",
        "T12",
        "T30",
6310     "T32",
        "T20",
        "T29",
        "T25",
        "T9",
6315     "T21",
        "T28",
        "T10",
        "T8",
        "T26",
6320     "T2",
        "T4",
        "T11",
        "T17",
        "T24"
6325     ],
    "energyUsed": 12394.375,
    "lenNotExecTasks": 14,
    "realTime": "3.959μs"
6330 },
{
    "step": 4,
    "lenTasksDone": 18,

```

```

6335         "tasksDone": [
            "T23",
            "T12",
            "T30",
            "T32",
            "T20",
            "T29",
6340         "T25",
            "T9",
            "T21",
            "T28",
            "T10",
6345         "T8",
            "T26",
            "T2",
            "T4",
            "T11",
6350         "T17",
            "T24"
        ],
        "energyUsed": 12394.375,
        "lenNotExecTasks": 14,
6355         "realTime": "6.492μs"
    },
    {
        "step": 5,
        "lenTasksDone": 18,
6360         "tasksDone": [
            "T23",
            "T12",
            "T30",
            "T32",
            "T20",
            "T29",
            "T25",
            "T9",
            "T21",
            "T28",
            "T10",
            "T8",
            "T26",
            "T2",
            "T4",
            "T11",
            "T17",
            "T24"
        ],
6370
        "energyUsed": 12394.375,
        "lenNotExecTasks": 14,
        "realTime": "4.145μs"
6375
    }
6380
]

```

```

6385 },
    Энергосберегающая": {
        "totalTasks": 32,
        "totalMachine": 8,
        "totalEnergyMachines": 12560,
6390 "allMachines": [
            {
                "id": "M1",
                "equipment": [
6395 1,
                3,
                9
            ],
            "energy": 1570.875,
            "AssignedTask": null
6400 },
            {
                "id": "M2",
                "equipment": [
6405 2,
                9
            ],
            "energy": 1570.875,
            "AssignedTask": null
        },
6410 {
            "id": "M3",
            "equipment": [
                1,
                7,
6415 14
            ],
            "energy": 1570.875,
            "AssignedTask": null
        },
6420 {
            "id": "M4",
            "equipment": [
                4,
                5,
6425 6,
                15
            ],
            "energy": 1570.875,
            "AssignedTask": null
6430 },
            {
                "id": "M5",
                "equipment": [
                11
6435 ],
            "energy": 1570.875,

```

```

        "AssignedTask": null
    },
    {
6440      "id": "M6",
        "equipment": [
            8,
            13
        ],
6445      "energy": 1570.875,
        "AssignedTask": null
    },
    {
6450      "id": "M7",
        "equipment": [
            10
        ],
        "energy": 1570.875,
        "AssignedTask": null
6455    },
    {
        "id": "M8",
        "equipment": [
6460            12
        ],
        "energy": 1570.875,
        "AssignedTask": null
    }
],
6465 "totalEnergyTasks": 25134,
    "allTasks": [
        {
            "id": "T23",
            "required_equipment": 8,
6470            "energy_cost": 1500,
            "CreatedAt": 1759044740
        },
        {
            "id": "T12",
6475            "required_equipment": 12,
            "energy_cost": 10,
            "CreatedAt": 1759044729
        },
        {
6480            "id": "T30",
            "required_equipment": 15,
            "energy_cost": 10,
            "CreatedAt": 1759044747
        },
6485    {
        "id": "T32",
        "required_equipment": 12,
        "energy_cost": 994,
    }
]

```

```

6490         "CreatedAt": 1759044719
        },
        {
        6495         "id": "T20",
            "required_equipment": 5,
            "energy_cost": 10,
            "CreatedAt": 1759044737
        },
        {
        6500         "id": "T29",
            "required_equipment": 14,
            "energy_cost": 1500,
            "CreatedAt": 1759044746
        },
        {
        6505         "id": "T25",
            "required_equipment": 10,
            "energy_cost": 1500,
            "CreatedAt": 1759044742
        },
        {
        6510         "id": "T9",
            "required_equipment": 9,
            "energy_cost": 1500,
            "CreatedAt": 1759044726
        },
        {
        6515         "id": "T21",
            "required_equipment": 6,
            "energy_cost": 1500,
            "CreatedAt": 1759044738
        6520     },
        {
            "id": "T28",
            "required_equipment": 13,
            "energy_cost": 10,
        6525         "CreatedAt": 1759044745
        },
        {
            "id": "T7",
            "required_equipment": 7,
        6530         "energy_cost": 1500,
            "CreatedAt": 1759044724
        },
        {
        6535         "id": "T1",
            "required_equipment": 1,
            "energy_cost": 1500,
            "CreatedAt": 1759044718
        },
        {
        6540         "id": "T10",

```



```

        "required_equipment": 10,
        "energy_cost": 10,
        "CreatedAt": 1759044727
    },
6545 {
        "id": "T8",
        "required_equipment": 8,
        "energy_cost": 10,
        "CreatedAt": 1759044725
6550 },
    {
        "id": "T14",
        "required_equipment": 14,
        "energy_cost": 10,
6555 "CreatedAt": 1759044731
    },
    {
        "id": "T18",
        "required_equipment": 3,
6560 "energy_cost": 10,
        "CreatedAt": 1759044735
    },
    {
        "id": "T26",
        "required_equipment": 11,
6565 "energy_cost": 10,
        "CreatedAt": 1759044743
    },
    {
6570 "id": "T2",
        "required_equipment": 2,
        "energy_cost": 10,
        "CreatedAt": 1759044719
    },
6575 {
        "id": "T4",
        "required_equipment": 4,
        "energy_cost": 10,
        "CreatedAt": 1759044721
6580 },
    {
        "id": "T3",
        "required_equipment": 3,
        "energy_cost": 1500,
6585 "CreatedAt": 1759044720
    },
    {
        "id": "T11",
        "required_equipment": 11,
6590 "energy_cost": 1500,
        "CreatedAt": 1759044728
    },

```

```

6595      {
        "id": "T16",
        "required_equipment": 1,
        "energy_cost": 10,
        "CreatedAt": 1759044733
      },
6600      {
        "id": "T22",
        "required_equipment": 7,
        "energy_cost": 10,
        "CreatedAt": 1759044739
      },
6605      {
        "id": "T17",
        "required_equipment": 2,
        "energy_cost": 1500,
        "CreatedAt": 1759044734
6610      },
      {
        "id": "T24",
        "required_equipment": 9,
        "energy_cost": 10,
6615      "CreatedAt": 1759044741
      },
      {
        "id": "T19",
        "required_equipment": 4,
6620      "energy_cost": 1500,
        "CreatedAt": 1759044736
      },
      {
        "id": "T27",
6625      "required_equipment": 12,
        "energy_cost": 1500,
        "CreatedAt": 1759044744
      },
      {
6630      "id": "T13",
        "required_equipment": 13,
        "energy_cost": 1500,
        "CreatedAt": 1759044730
      },
6635      {
        "id": "T31",
        "required_equipment": 7,
        "energy_cost": 1490,
        "CreatedAt": 1759044718
6640      },
      {
        "id": "T15",
        "required_equipment": 15,
        "energy_cost": 1500,

```

```

6645         "CreatedAt": 1759044732
        },
        {
        6650         "id": "T5",
         "required_equipment": 5,
         "energy_cost": 1500,
         "CreatedAt": 1759044722
        },
        {
        6655         "id": "T6",
         "required_equipment": 6,
         "energy_cost": 10,
         "CreatedAt": 1759044723
        }
    ],
    6660 "totalTime": "39.143µs",
    "stepMetrics": [
        {
        6665         "step": 1,
         "lenTasksDone": 23,
         "tasksDone": [
            "T18",
            "T2",
            "T14",
            "T30",
        6670         "T26",
            "T28",
            "T10",
            "T12",
            "T16",
        6675         "T24",
            "T22",
            "T20",
            "T11",
            "T8",
        6680         "T25",
            "T32",
            "T9",
            "T17",
            "T31",
        6685         "T4",
            "T23",
            "T6",
            "T21"
        ],
        6690         "energyUsed": 12394.375,
         "lenNotExecTasks": 9,
         "realTime": "13.353µs"
        },
        {
        6695         "step": 2,
         "lenTasksDone": 23,

```

```

6700     "tasksDone": [
        "T18",
        "T2",
        "T14",
        "T30",
        "T26",
        "T28",
        "T10",
6705     "T12",
        "T16",
        "T24",
        "T22",
        "T20",
6710     "T11",
        "T8",
        "T25",
        "T32",
        "T9",
6715     "T17",
        "T31",
        "T4",
        "T23",
        "T6",
6720     "T21"
    ],
    "energyUsed": 12394.375,
    "lenNotExecTasks": 9,
    "realTime": "6.512μs"
6725 },
    {
        "step": 3,
        "lenTasksDone": 23,
        "tasksDone": [
6730     "T18",
        "T2",
        "T14",
        "T30",
        "T26",
6735     "T28",
        "T10",
        "T12",
        "T16",
        "T24",
6740     "T22",
        "T20",
        "T11",
        "T8",
        "T25",
6745     "T32",
        "T9",
        "T17",
        "T31",

```

```

6750         "T4",
            "T23",
            "T6",
            "T21"
        ],
        "energyUsed": 12394.375,
6755     "lenNotExecTasks": 9,
        "realTime": "6.179µs"
    },
    {
6760     "step": 4,
        "lenTasksDone": 23,
        "tasksDone": [
            "T18",
            "T2",
            "T14",
6765     "T30",
            "T26",
            "T28",
            "T10",
            "T12",
6770     "T16",
            "T24",
            "T22",
            "T20",
            "T11",
6775     "T8",
            "T25",
            "T32",
            "T9",
            "T17",
6780     "T31",
            "T4",
            "T23",
            "T6",
            "T21"
        ],
6785     "energyUsed": 12394.375,
        "lenNotExecTasks": 9,
        "realTime": "4.595µs"
    },
6790     {
        "step": 5,
        "lenTasksDone": 23,
        "tasksDone": [
            "T18",
6795     "T2",
            "T14",
            "T30",
            "T26",
            "T28",
6800     "T10",

```

```

        "T12",
        "T16",
        "T24",
        "T22",
6805    "T20",
        "T11",
        "T8",
        "T25",
        "T32",
6810    "T9",
        "T17",
        "T31",
        "T4",
        "T23",
6815    "T6",
        "T21"
    ],
    "energyUsed": 12394.375,
    "lenNotExecTasks": 9,
    "realTime": "4.257µs"
6820 }
    }
6825 }

```

А.2 Вывод csv

Листинг 2: файл summaryresults.csv

```

СценарийСтратегияВыполнено
, , задачИспользовано, энергииВсего, энергии,% выполненияСр,. энергия на
задачуОбщее, время (µs)
equalПрямого, перебора,25,19788.00,25424.00,78.1%,791.52,36.216
equalРыночных, методов,25,19788.00,25424.00,78.1%,791.52,55.478
5 equalЭнергосберегающая, ,28,19788.00,25424.00,87.5%,706.71,52.016
equal,RL (Q-learning),26,18446.00,25424.00,81.2%,709.46,226.524
equalМинимизация, Вр3,25,19788.00,25424.00,78.1%,791.52,113.445
max,RL (Q-learning),30,22966.00,50848.00,93.8%,765.53,250.107
maxМинимизация, Вр3,29,22966.00,50848.00,90.6%,791.93,151.103
10 maxПрямого, перебора,29,22966.00,50848.00,90.6%,791.93,28.875
maxРыночных, методов,29,22966.00,50848.00,90.6%,791.93,81.775
maxЭнергосберегающая, ,30,22966.00,50848.00,93.8%,765.53,77.164
minЭнергосберегающая, ,23,12475.00,12712.00,71.9%,542.39,60.637
min,RL (Q-learning),21,12406.00,12712.00,65.6%,590.76,236.423
15 minМинимизация, Вр3,18,12475.00,12712.00,56.2%,693.06,107.655
minПрямого, перебора,18,12475.00,12712.00,56.2%,693.06,35.765
minРыночных, методов,18,12475.00,12712.00,56.2%,693.06,53.883

```